AWS Lab | Storage and Database Services Lab

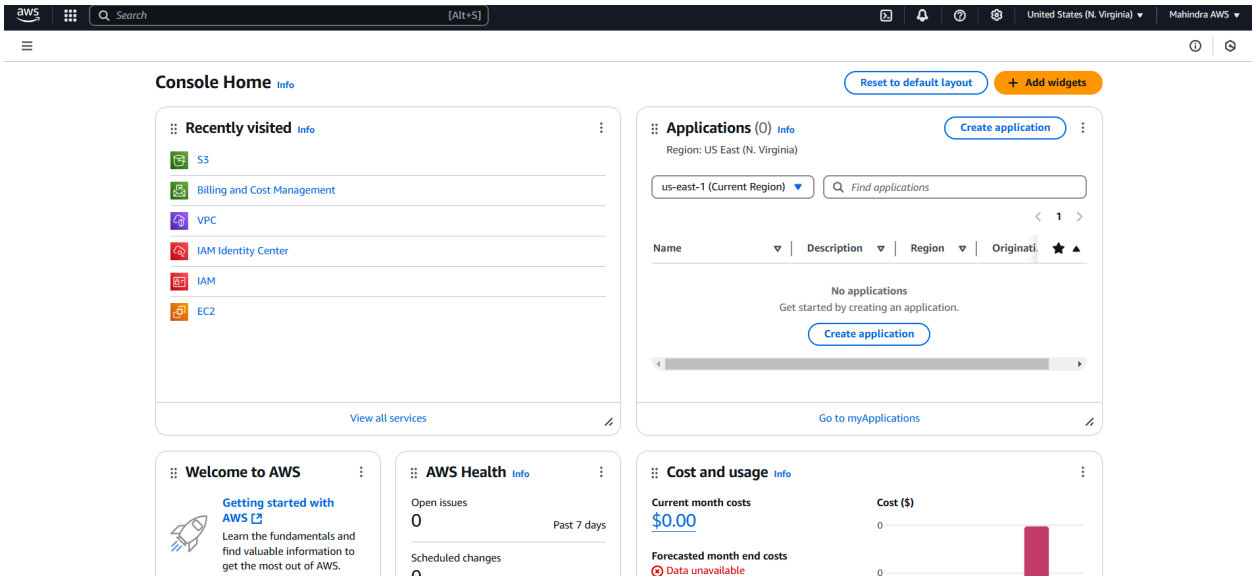
Author : Mahindra Guptha Kotha
Mail ID: mkotha@smu.edu

EXERCISE 3.1

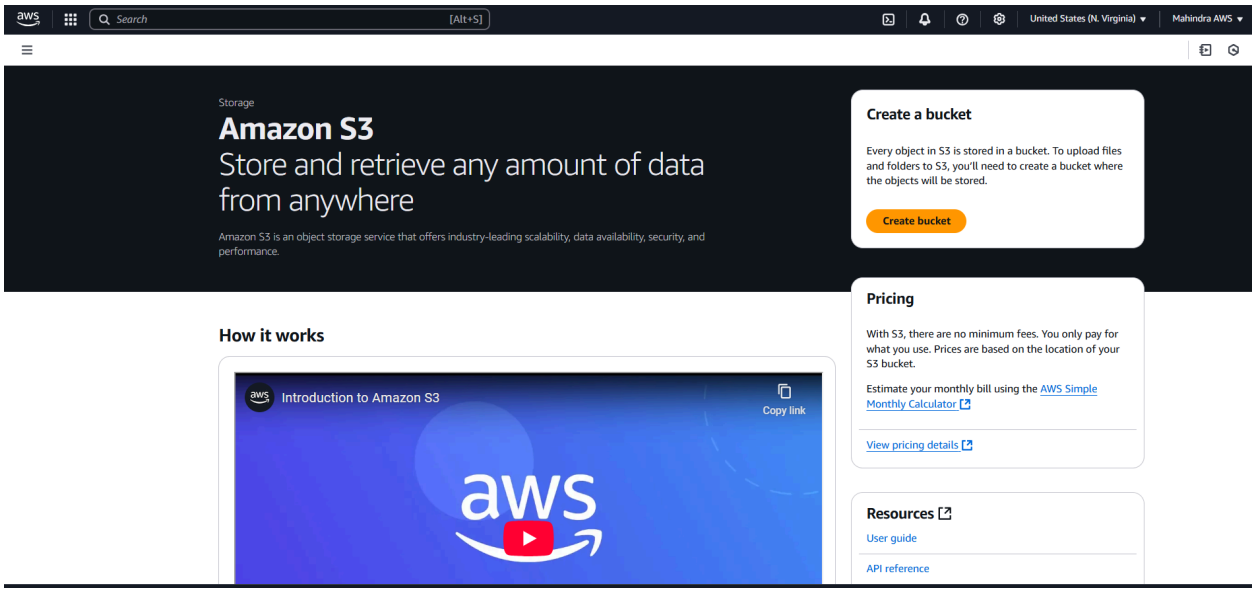
Create a New S3 Bucket and Upload a File

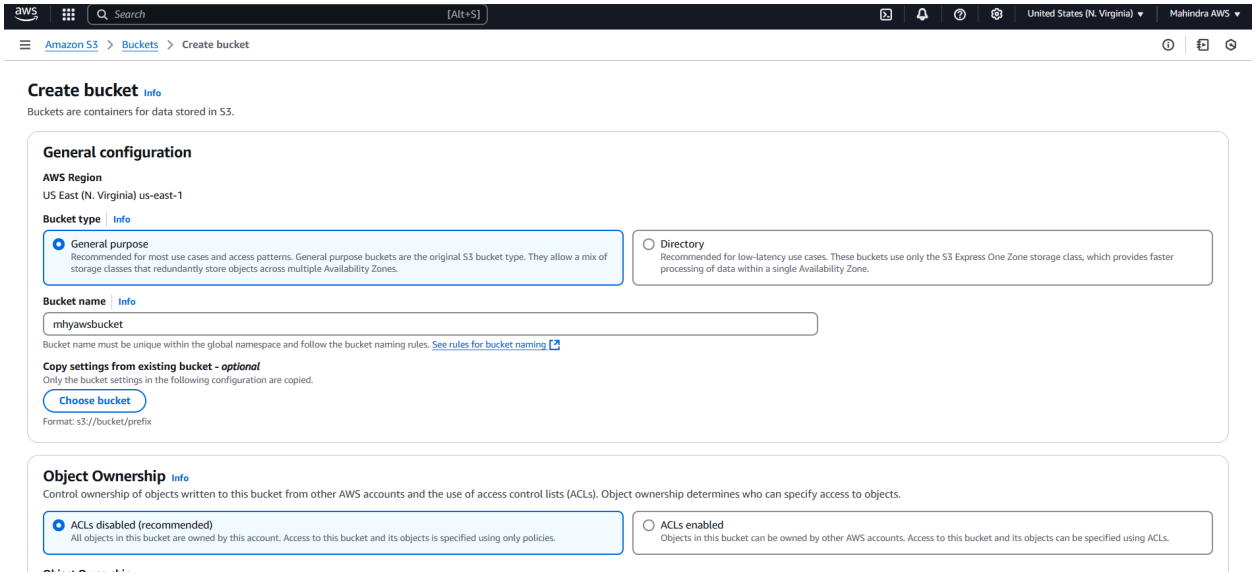
1. From the AWS S3 dashboard, create a new bucket and use either the dashboard or the AWS CLI (`aws s3 cp mylocalfile.txt s3://mybucketname/`) to upload one or more files, giving public read access to the object.

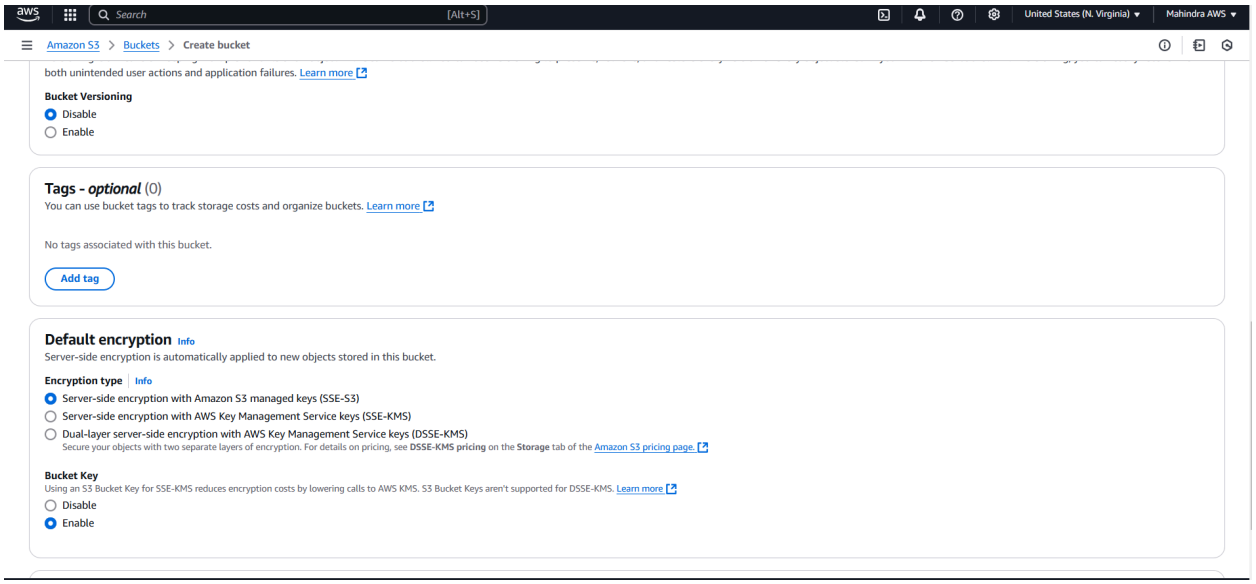
Sign in as a root account.

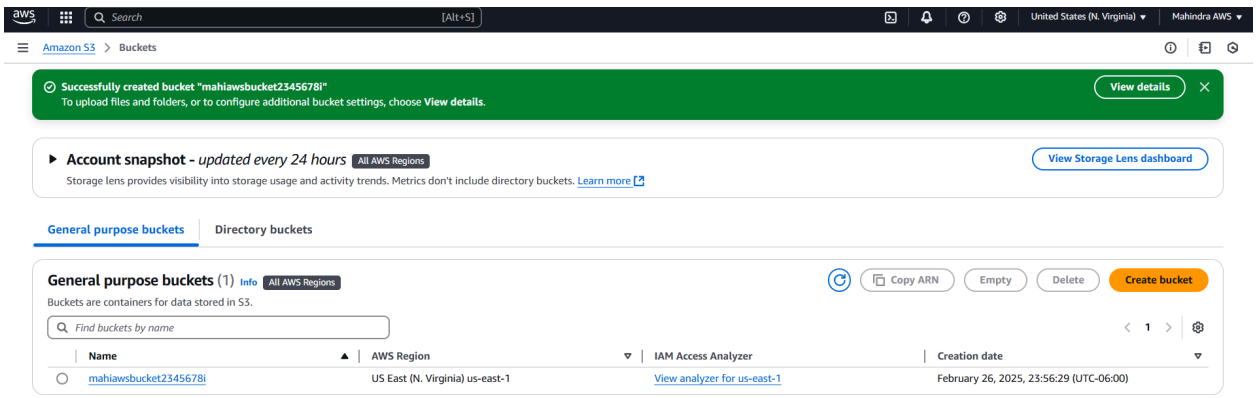


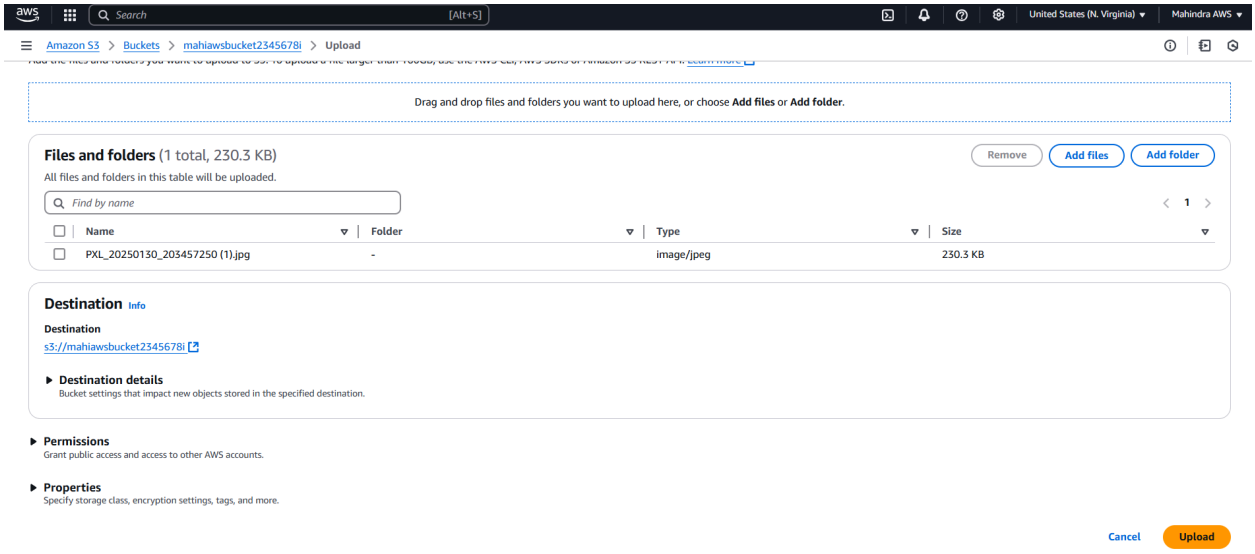
Select S3 and create a bucket.

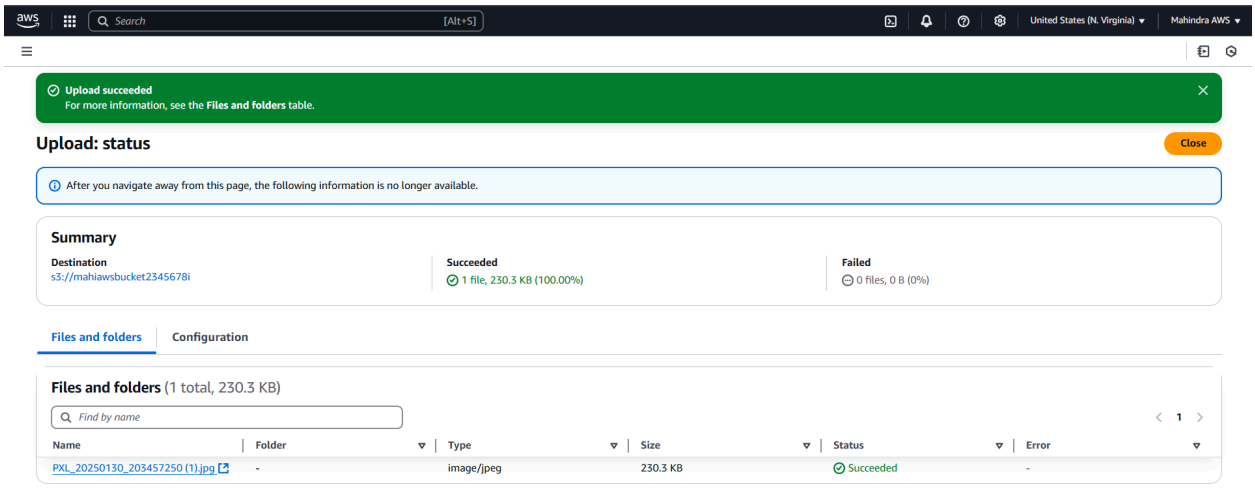



Search [Alt+S]
United States (N. Virginia)
Mahindra AWS
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket name Info
mhyawsbucket
Bucket name must be unique within the global namespace and follow the bucket naming rules. See rules for bucket naming
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

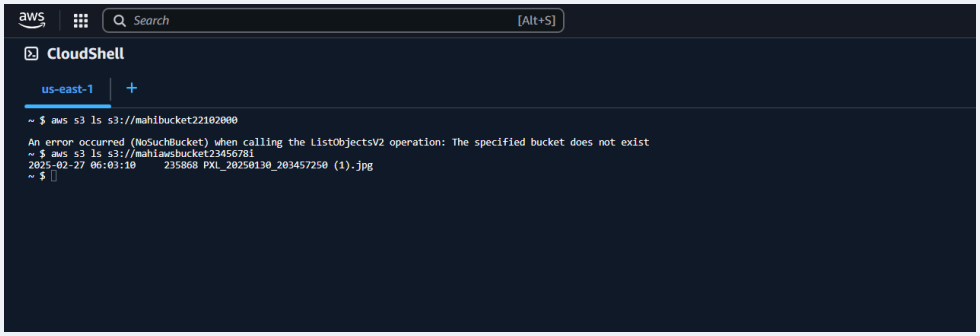

Search [Alt+S]
United States (N. Virginia)
Mahindra AWS
Amazon S3 > Buckets > Create bucket
both unintended user actions and application failures. Learn more
Bucket Versioning
Disable
Enable
Tags - optional (0)
You can use bucket tags to track storage costs and organize buckets. Learn more
No tags associated with this bucket.
Add tag
Default encryption Info
Server-side encryption is automatically applied to new objects stored in this bucket.
Encryption type Info
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable


Search [Alt+S]
United States (N. Virginia)
Mahindra AWS
Amazon S3 > Buckets
Successfully created bucket "mahiawsbucket2345678i"
To upload files and folders, or to configure additional bucket settings, choose View details.
View details
Account snapshot - updated every 24 hours
All AWS Regions
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. Learn more
View Storage Lens dashboard
General purpose buckets
Directory buckets
General purpose buckets (1) Info
All AWS Regions
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
Name
AWS Region
IAM Access Analyzer
Creation date
mahiawsbucket2345678i
US East (N. Virginia) us-east-1
View analyzer for us-east-1
February 26, 2025, 23:56:29 (UTC-06:00)





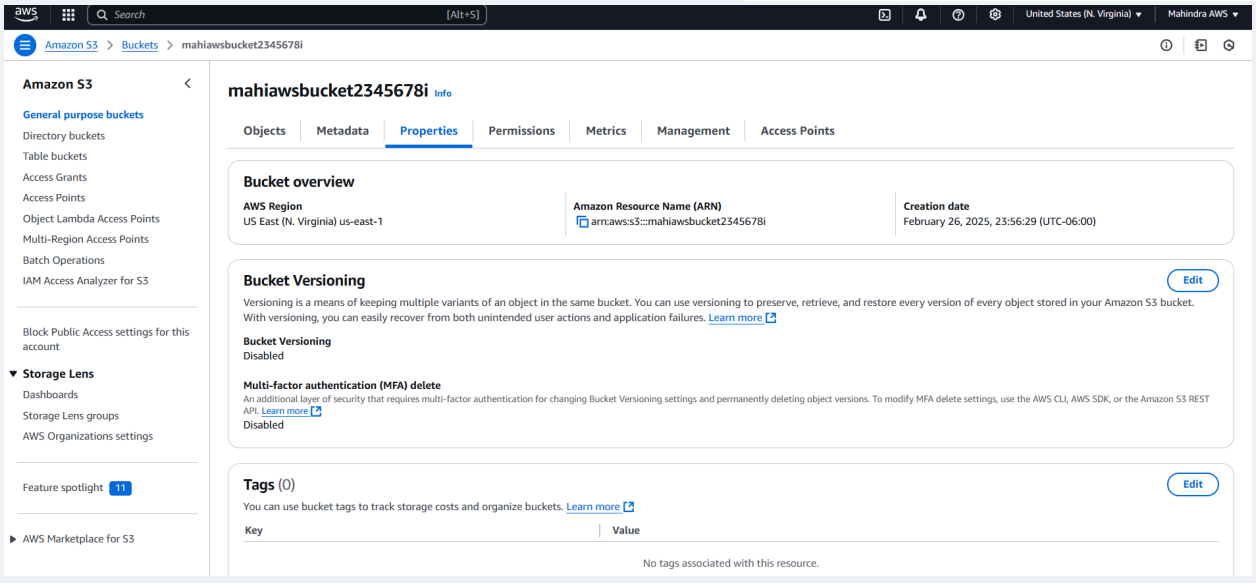
2. Use the AWS CLI to list the contents of your new bucket to confirm that the new file or files are there:
   `aws s3 ls s3://mahiawsbucket2345678i`



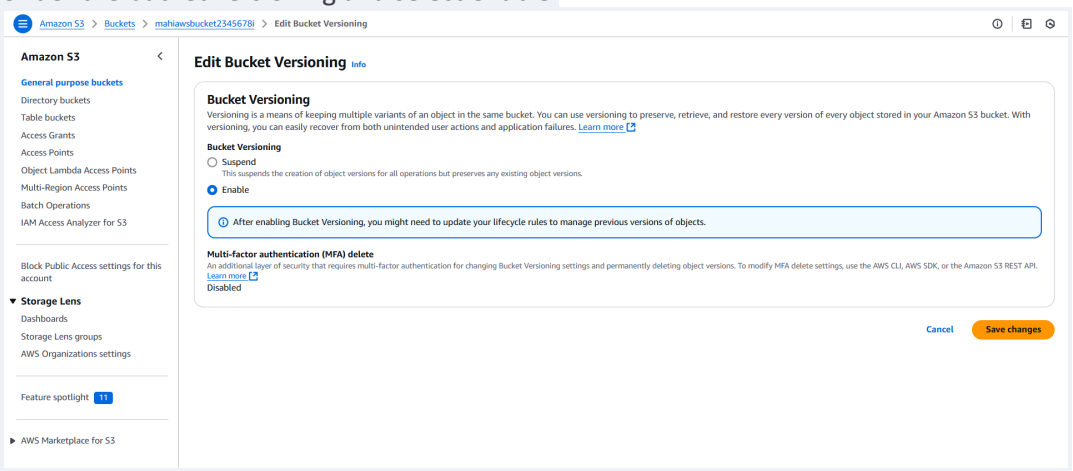
EXERCISE 3.2

ENABLE VERSIONING AND LIFE CYCLE MANAGEMENT FOR AN S3 BUCKET

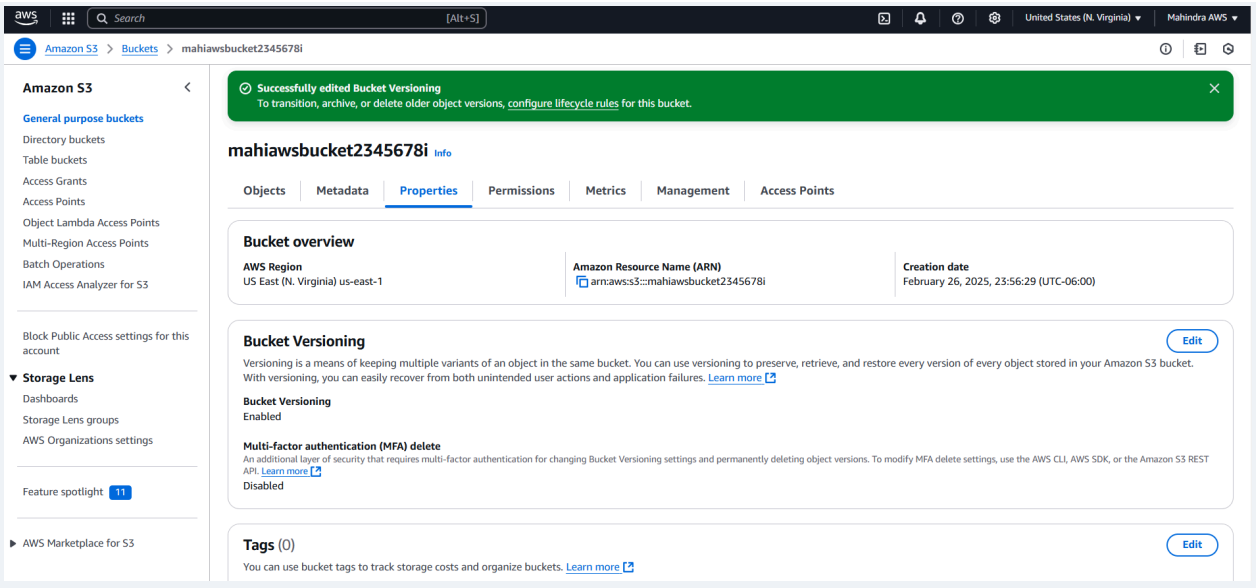
Select your bucket and edit its properties to enable versioning.



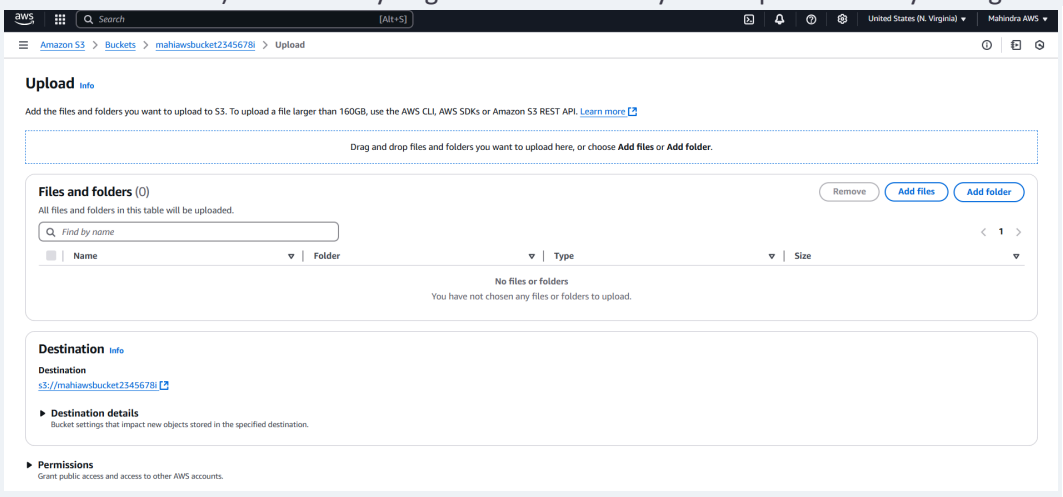
Under the bucket versioning and select enable.

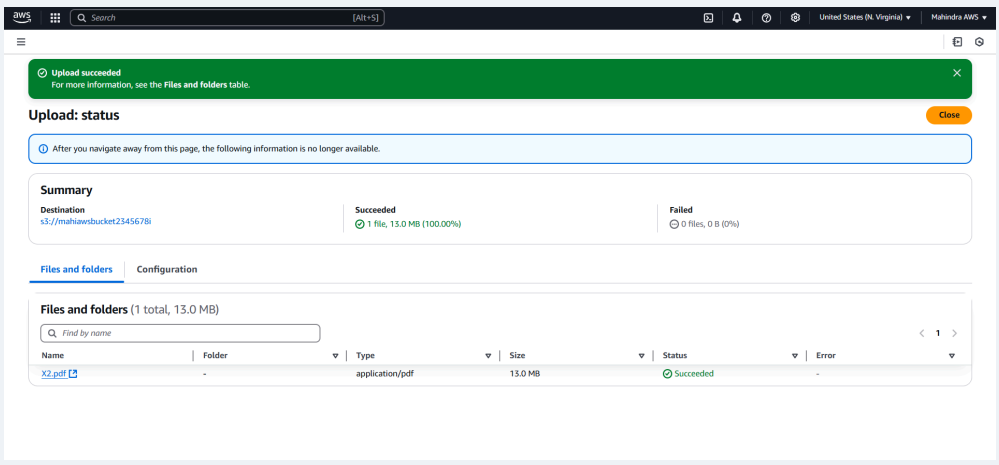


Click on save changes.

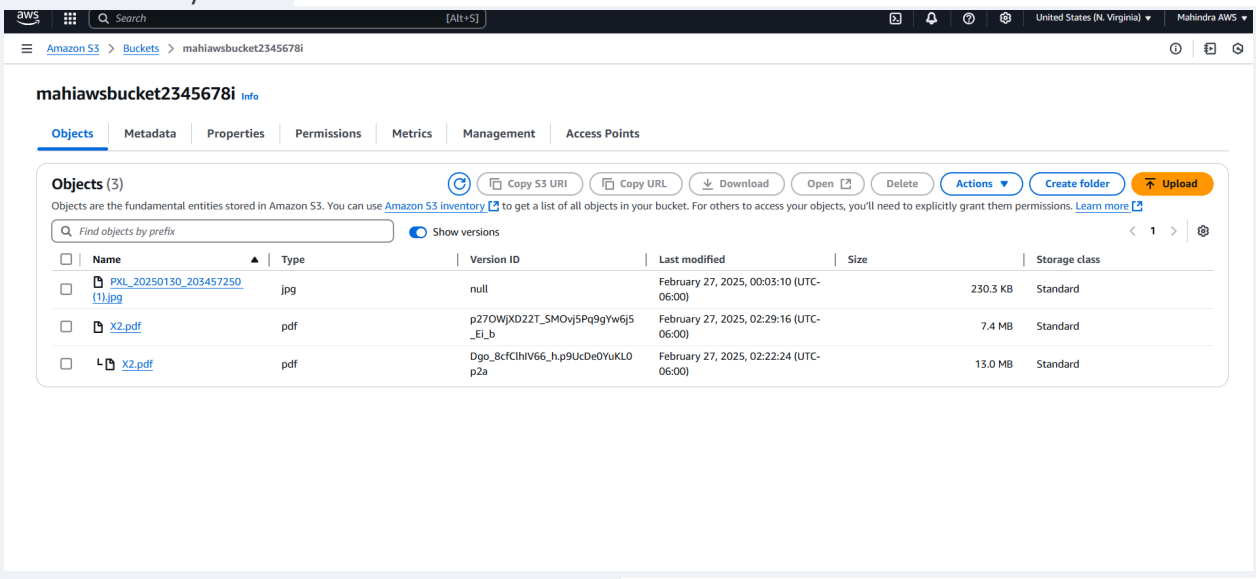


Upload a file to that bucket, edit the copy on your local computer, and upload the new copy (keeping the filename the same). Make sure you give the new file any access permissions you might need.

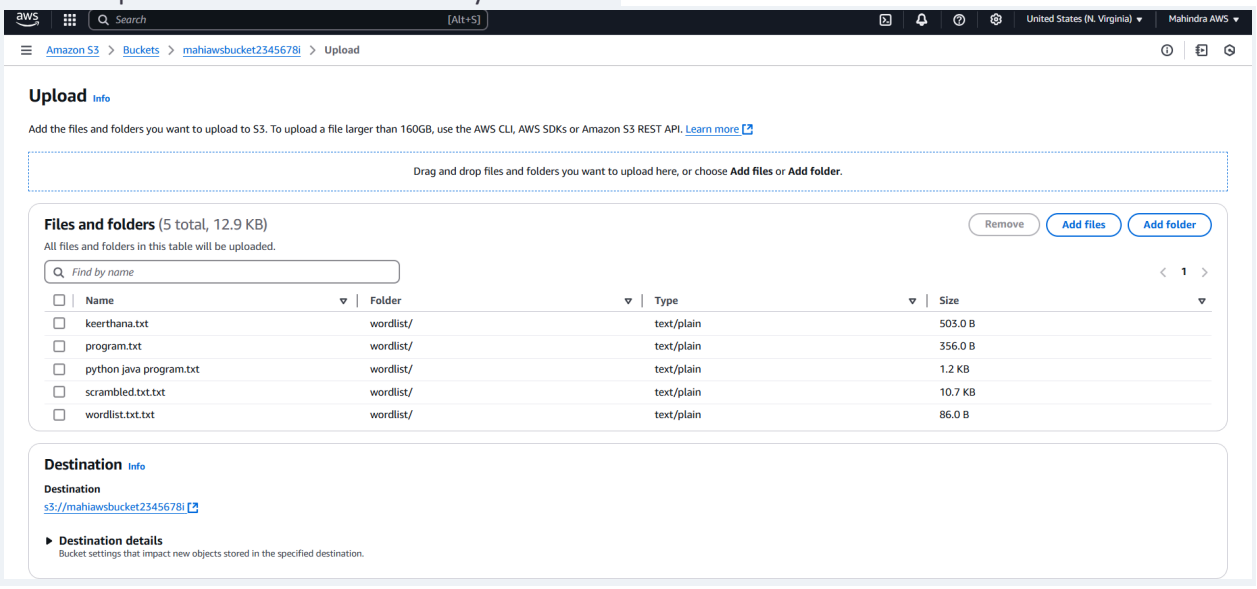


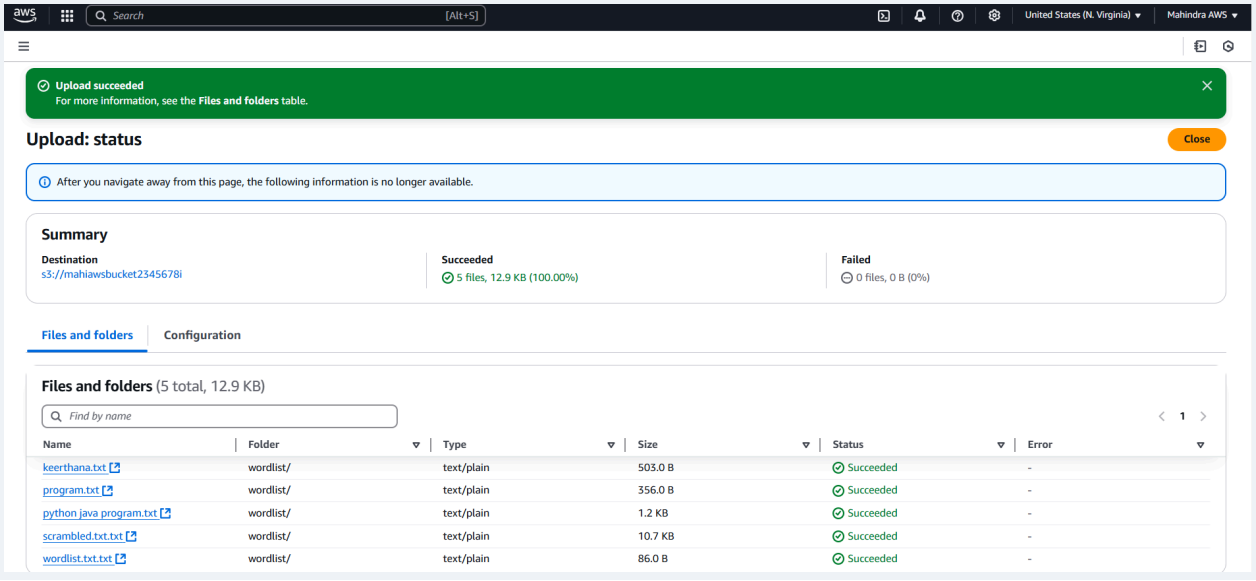


With the contents of the bucket displayed in the dashboard, select Show Versions. You should now see two versions of your file.

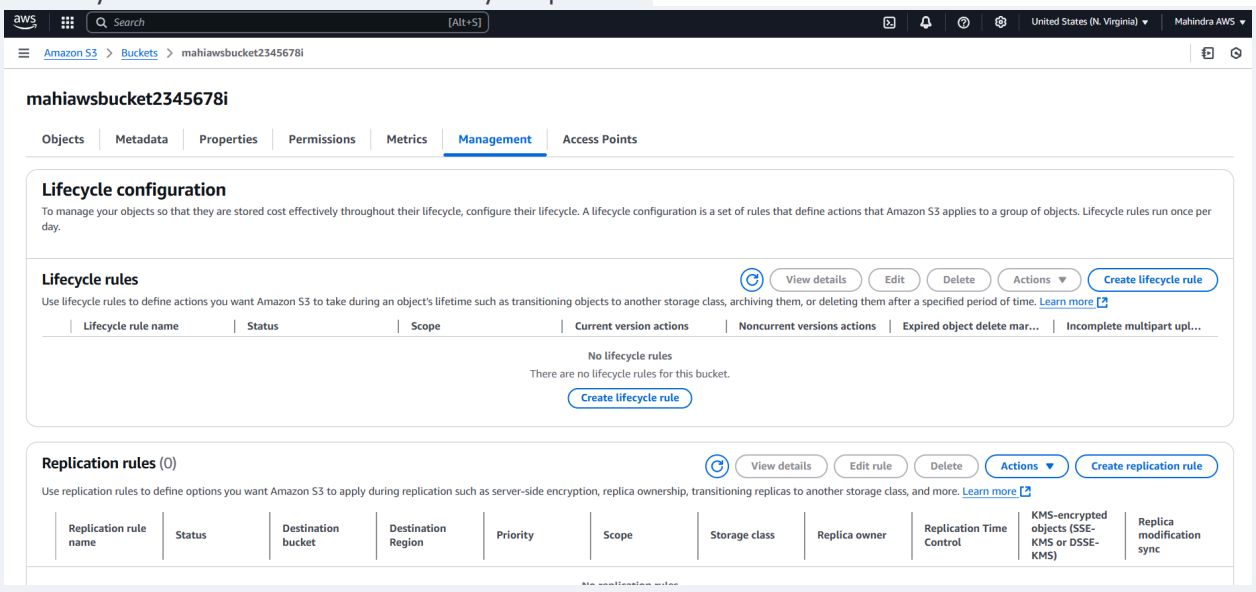


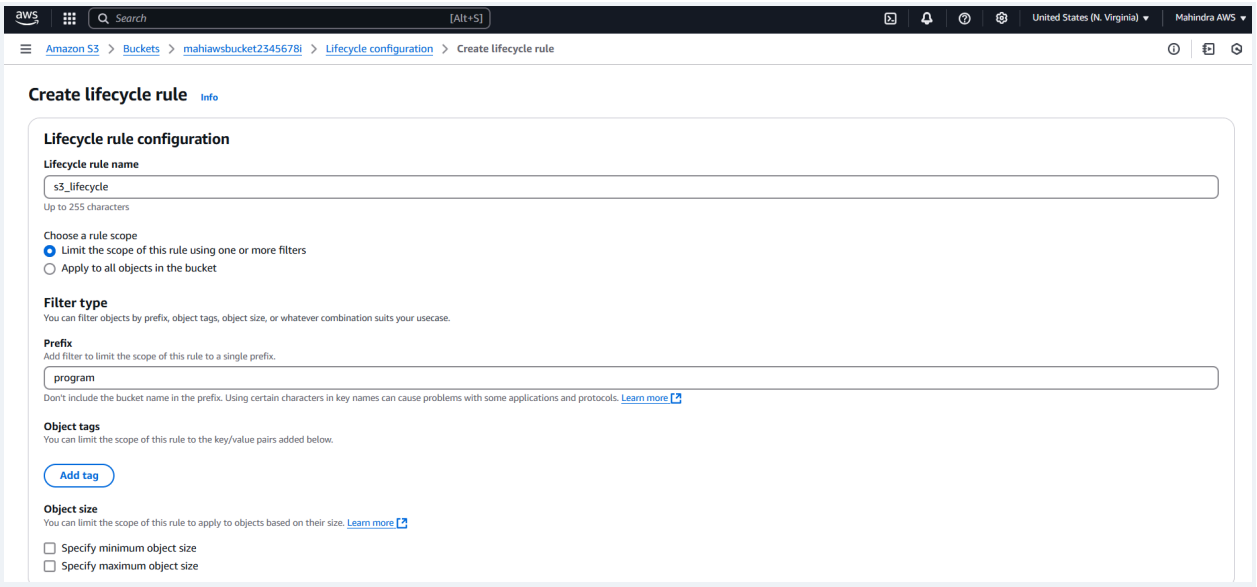
Add a couple of directories with files to your bucket.



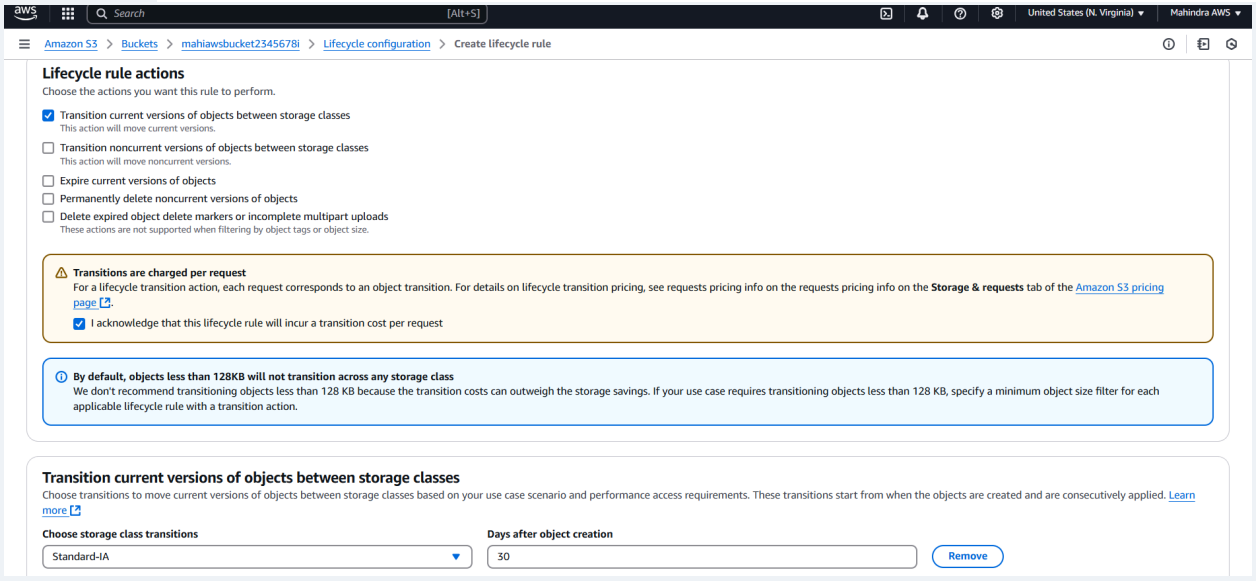


On the bucket's Management tab, create a life cycle rule and specify a prefix/tag filter that matches the directory name of one of the directories you uploaded.

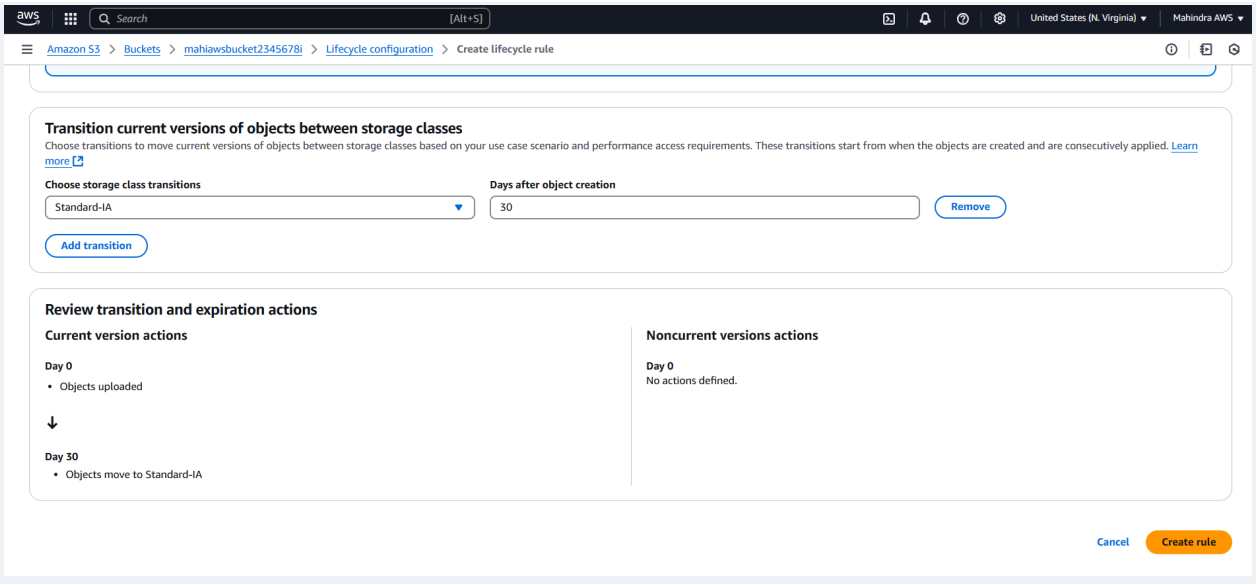


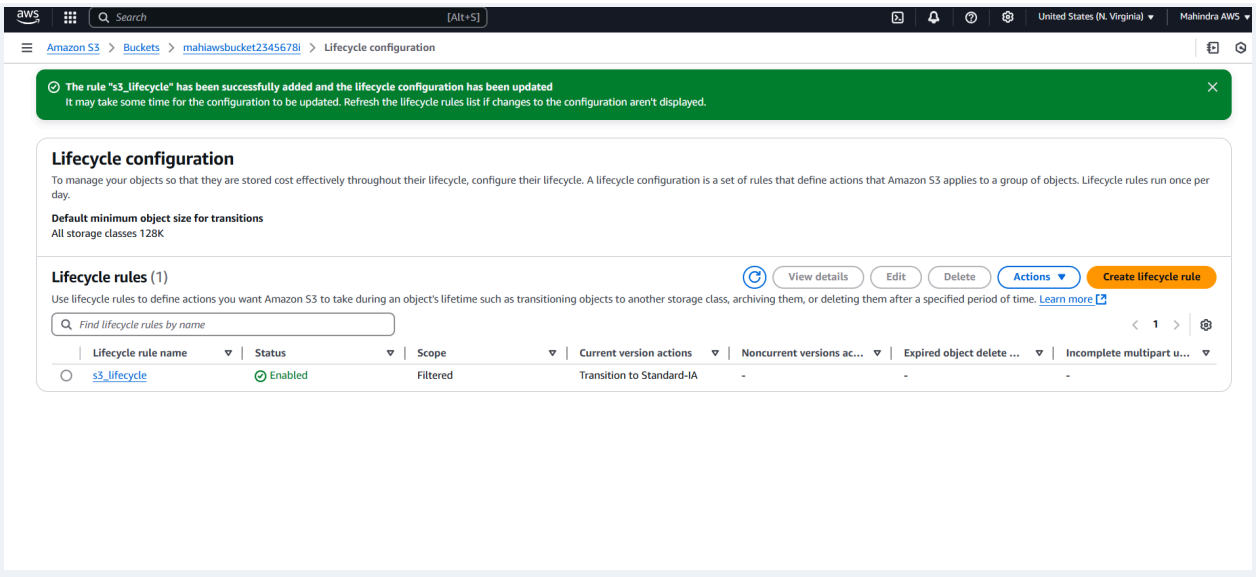


Configure a life cycle rule by adding transitions and configuring the transition timing (in days) and target for each one.



You'll need to be patient to test this configuration because the minimum lag between transitions is 30 days.

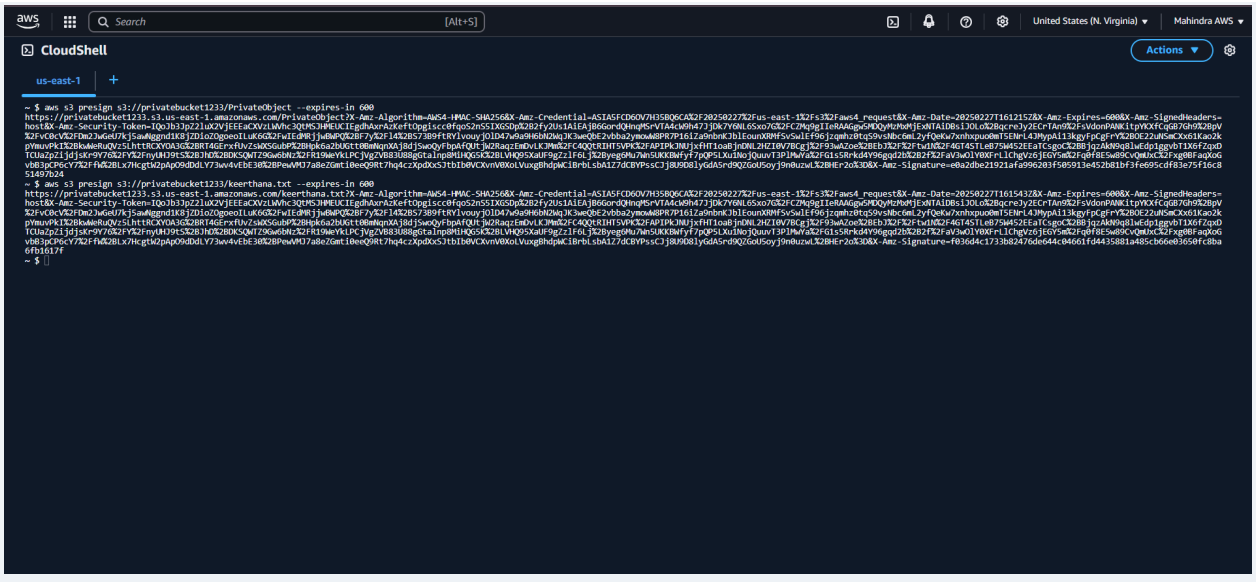




EXERCISE 3.3

Generate and Use a Presigned URL

1. Use the complete URL of a private object in an S3 bucket belonging to you to generate a presigned URL using a variation of this command:
   `aws s3 presign s3://privatebucket1233/keerthana.txt --expires-in 600`



2. Copy the full URL from the command output, and from a browser that's not logged into your AWS account, try to open the file.

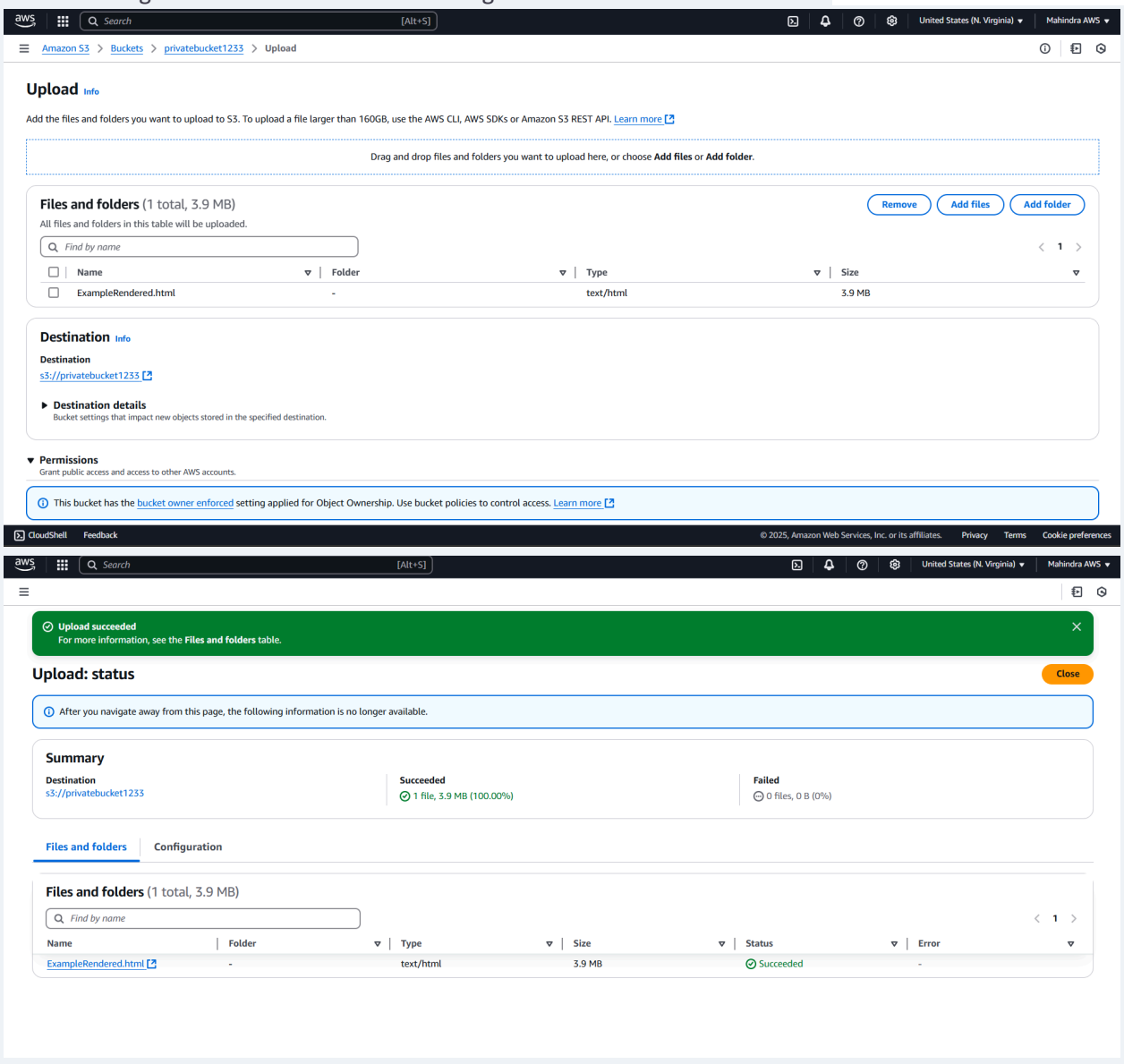
```
fav dish  -  nathu kodi, fish
hero      -   from different people
place - water and greenery like maldivies
fear  -  public speaking, dogs
angry - Work dealing lo
change yourself -  sleeping
motivates to work hard - doing best with work
best way to learn things - physical ga chusi nerchukovali
laugh  -  perfected jokes
nickname  -   keetoo, keera, keerthi, sunny, sai
music  -   instrumental music
yourself in five years -  simple lifecoffe
coffe or tea - tea
daring    -
clean
```

3. Wait for the URL to expire and try again. This time, it should not work.

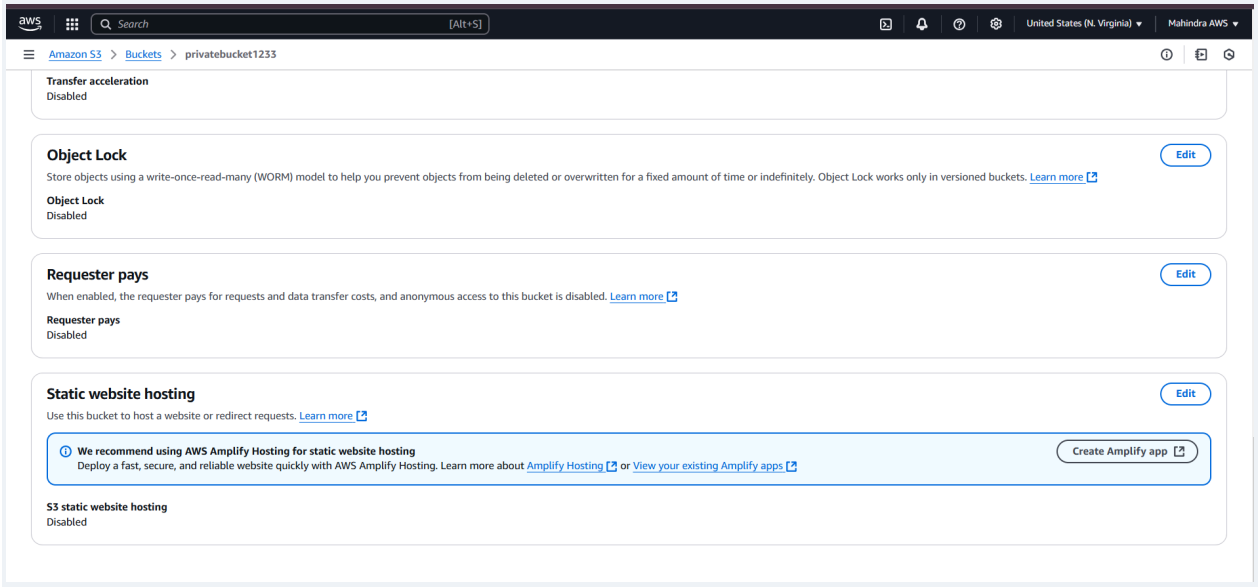
EXERCISE 3.4

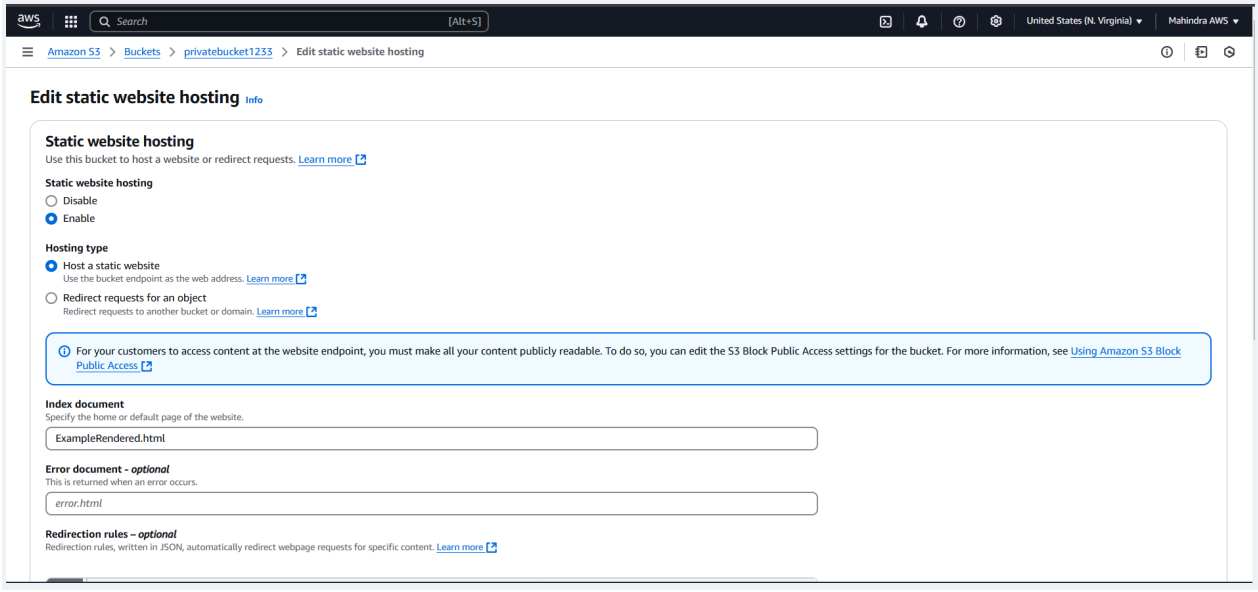
ENABLE STATIC WEBSITE HOSTING FOR AN S3 BUCKET

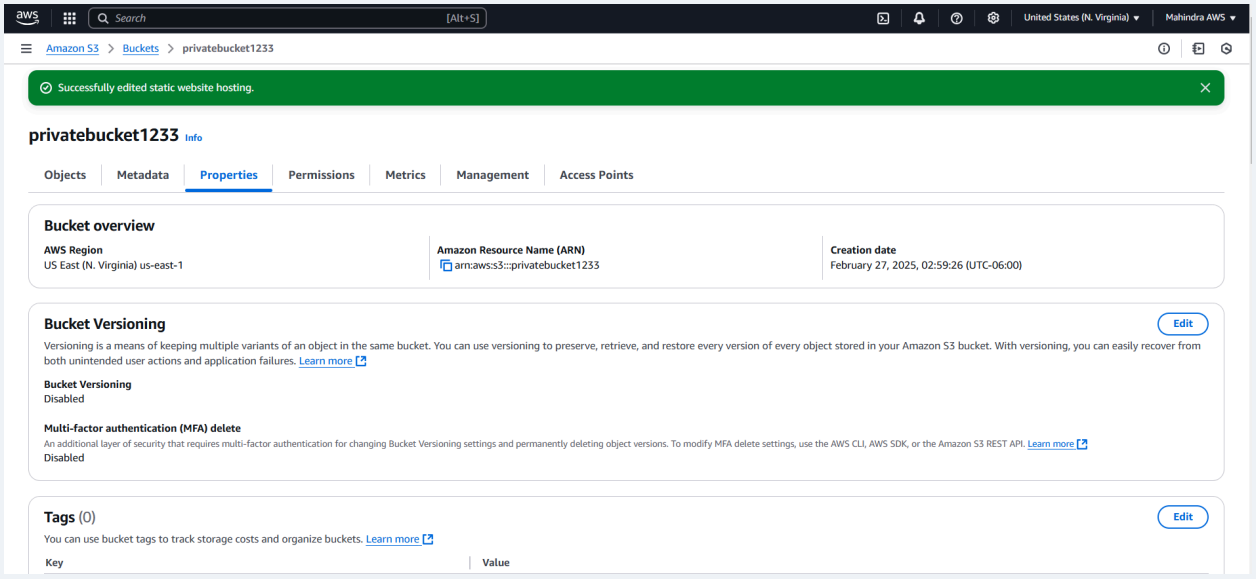
From the S3 dashboard, select (or create) an S3 bucket that contains at least one file with some simple text named index.html. Any files you want to be accessible should be readable by the public. You may need to change the Block Public Access settings for the bucket to allow access.



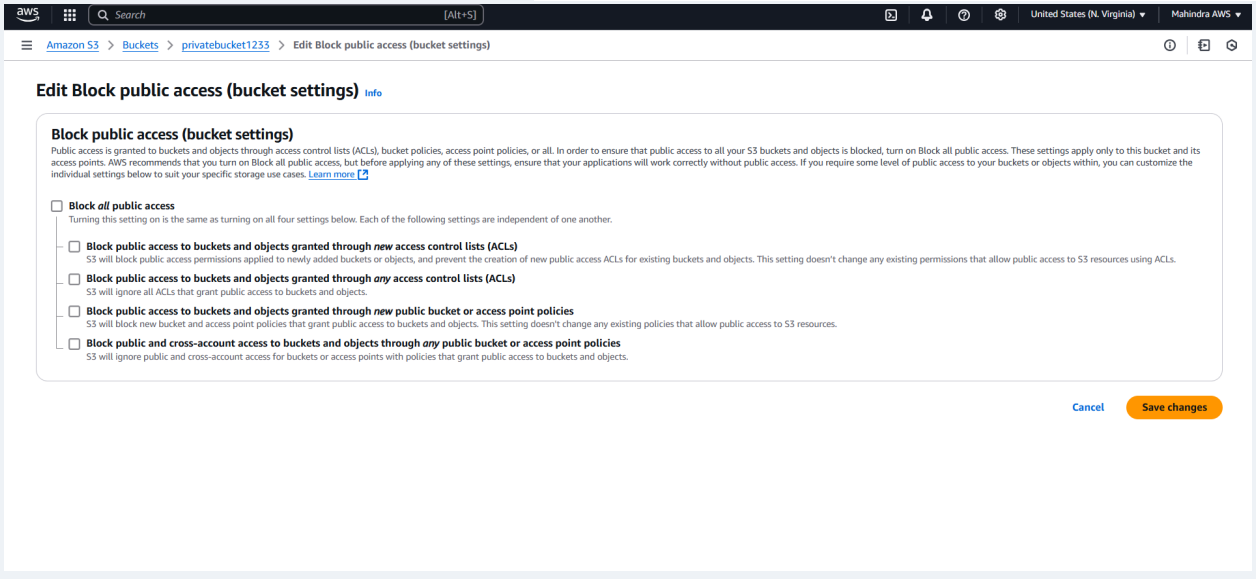
On the bucket's Properties tab, enable static website hosting and specify your index.html file as your index document.


Search
[Alt+S]
United States (N. Virginia)
Mahindra AWS
Amazon S3 > Buckets > privatebucket1233
Transfer acceleration
Disabled
Object Lock
Edit
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more
Object Lock
Disabled
Requester pays
Edit
When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more
Requester pays
Disabled
Static website hosting
Edit
Use this bucket to host a website or redirect requests. Learn more
We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps
Create Amplify app
S3 static website hosting
Disabled

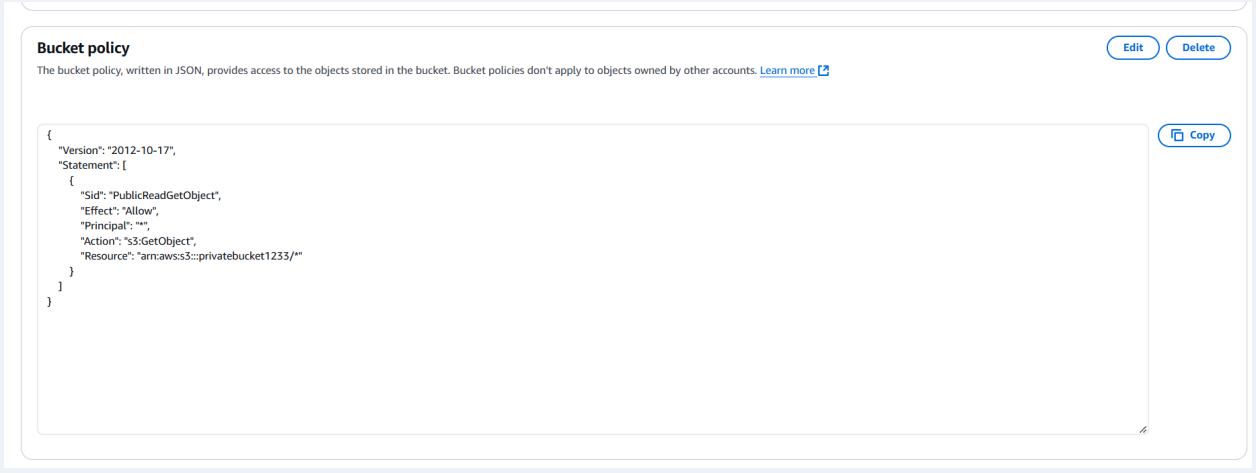

Search
[Alt+S]
United States (N. Virginia)
Mahindra AWS
Amazon S3 > Buckets > privatebucket1233 > Edit static website hosting
Edit static website hosting
Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
ExampleRendered.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules – optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more



Paste the static website endpoint into the URL field of a browser that's not logged into your AWS account and confirm that you can access the website.

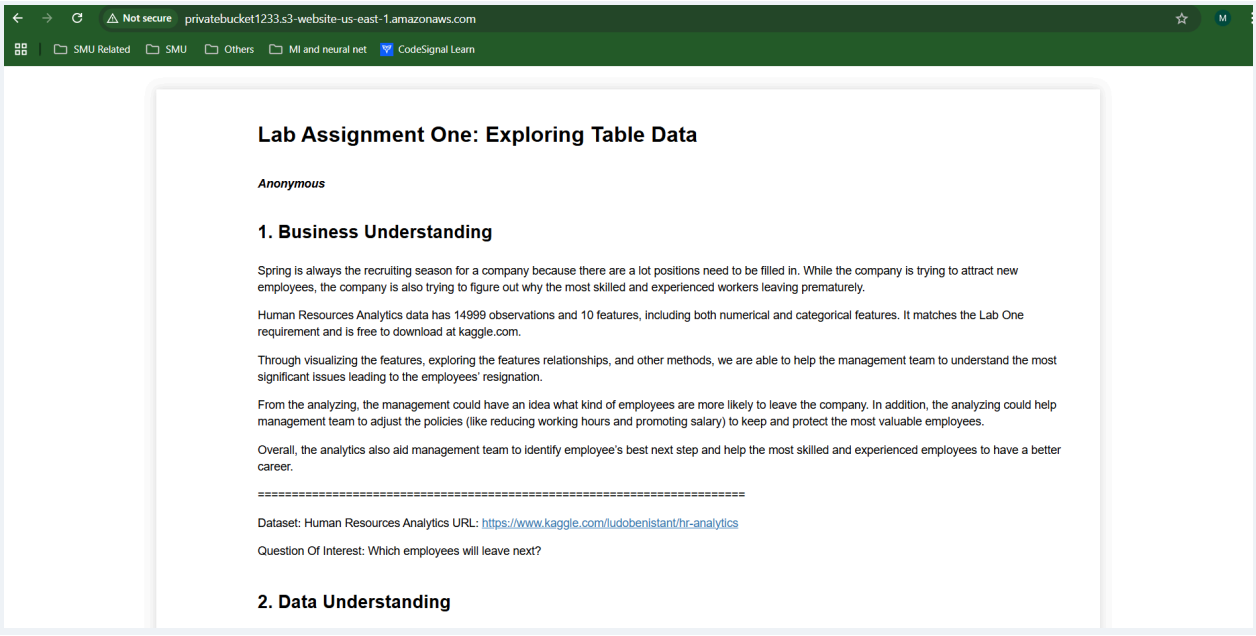


Go to bucket and permission and then bucket policy and add this json code



Note that you can also enable static website hosting from the AWS CLI using a variation of these two commands:

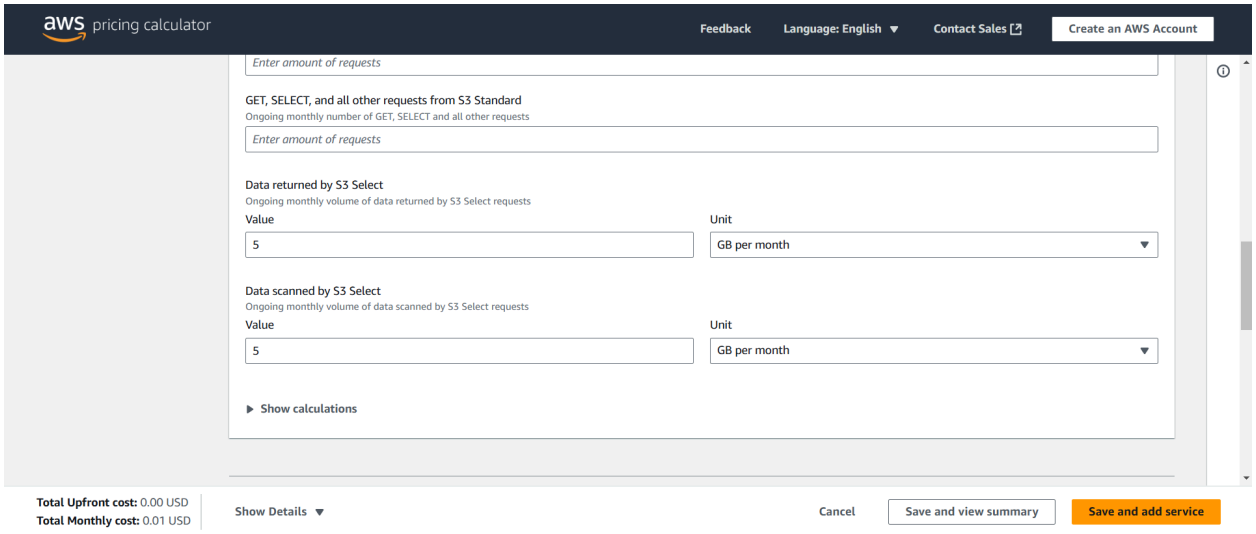
```
aws s3api put-bucket-acl --bucket my-bucket --acl public-read
aws s3 website s3://my-bucket/ --index-document index.html\
--error-document error.html
```



EXERCISE 3.5

Calculate the Total Life Cycle Costs for Your Data

Use the AWS Simple Monthly Calculator (http://calculator.s3.amazonaws.com/index.html) to estimate the total monthly costs of the scenario described at the beginning of this section. Even better, use a scenario that fits your own business. Try to include a full usage scenario, including requests, scans, and data retrieval. Note that you access the S3 part of the calculator by clicking the Amazon S3 tab on the left, and you can keep track of your itemized estimate using the Estimate Of Your Monthly Bill tab along the top.



Note that AWS has been trying to deprecate the Simple Monthly Calculator for some time in favor of their new AWS Pricing Calculator (https://calculator.aws/#). However, since so many customers have communicated their preference for the old tool, AWS continues to extend its life. Nevertheless, be aware that the move will likely happen eventually.

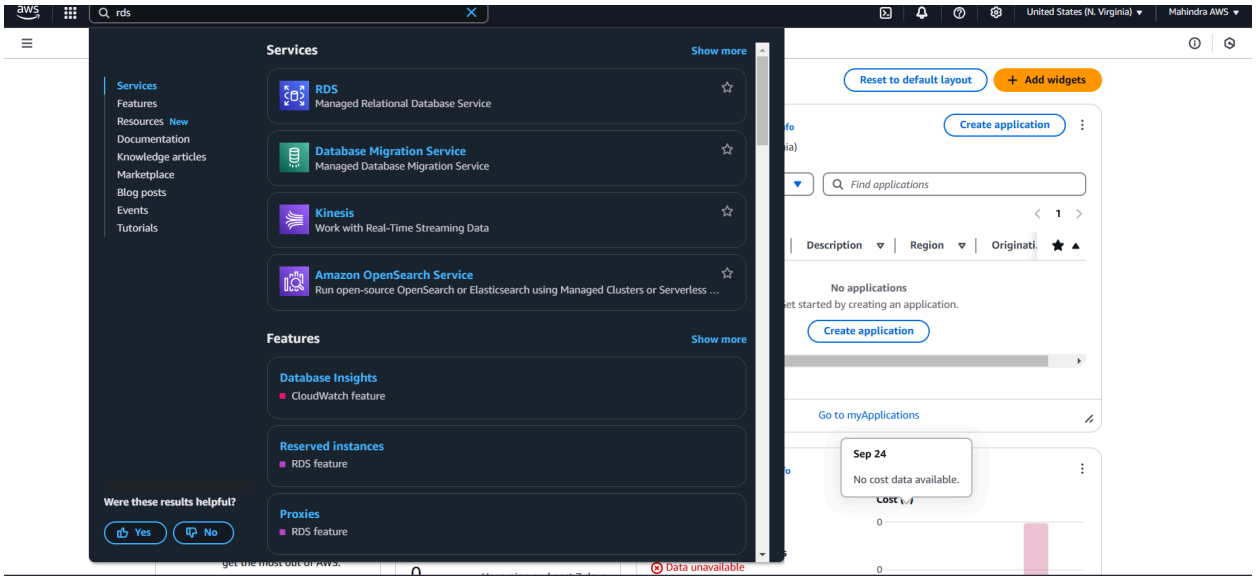
Chapter 5

EXERCISE 5.1

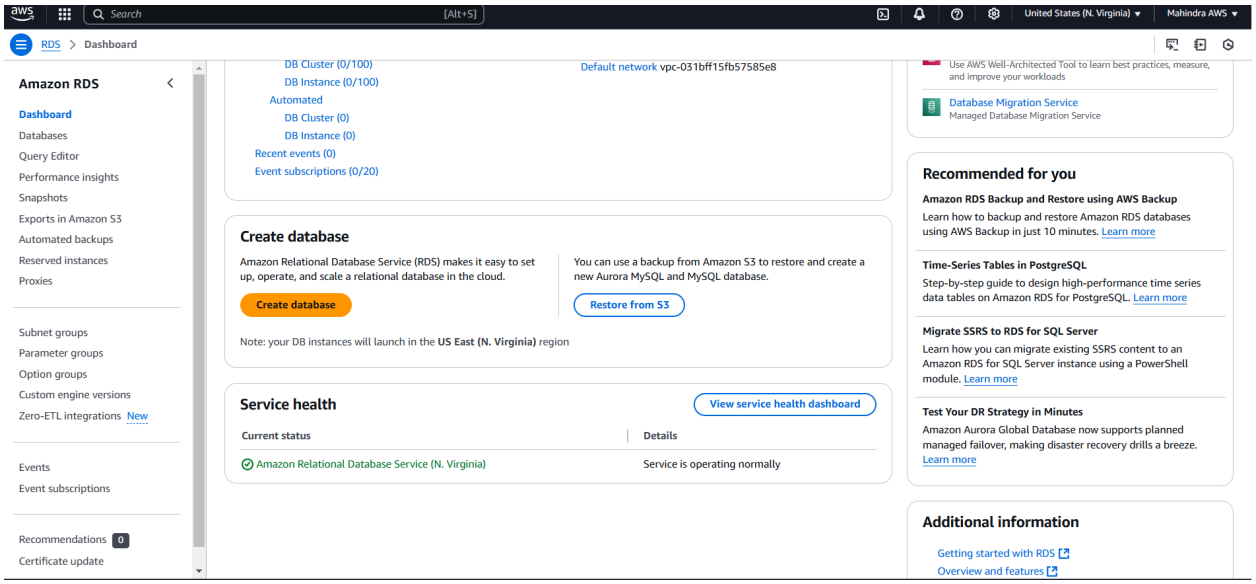
Create an RDS Database Instance

In this exercise, you'll create an RDS database instance using MariaDB as the database engine. Free Tier accounts won't incur any charges.

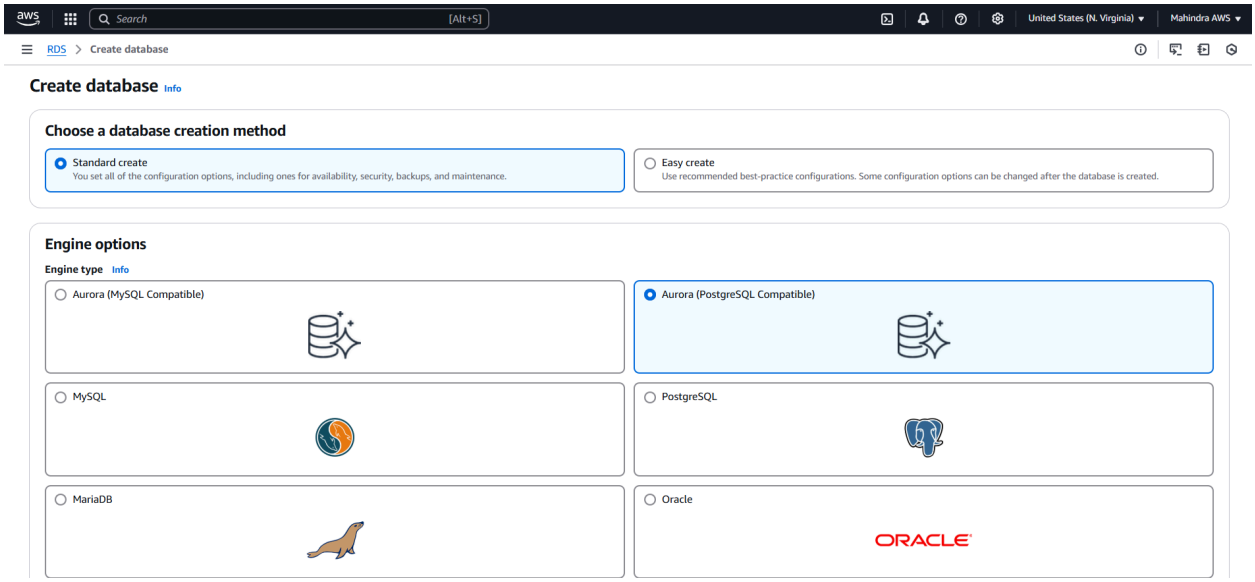
1. Click the RDS Dashboard link in the RDS service console.



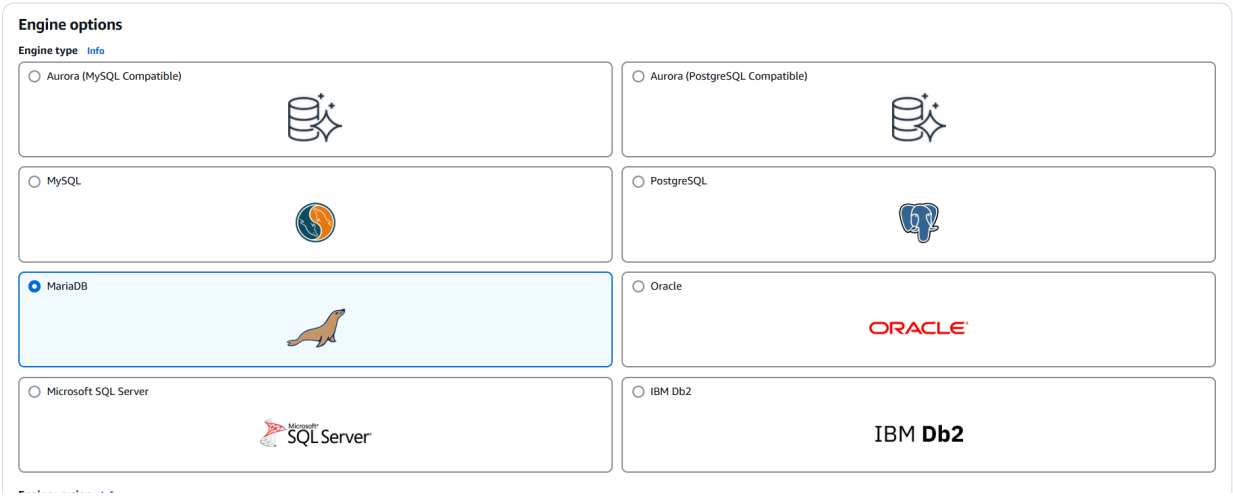
2. Click Create Database.



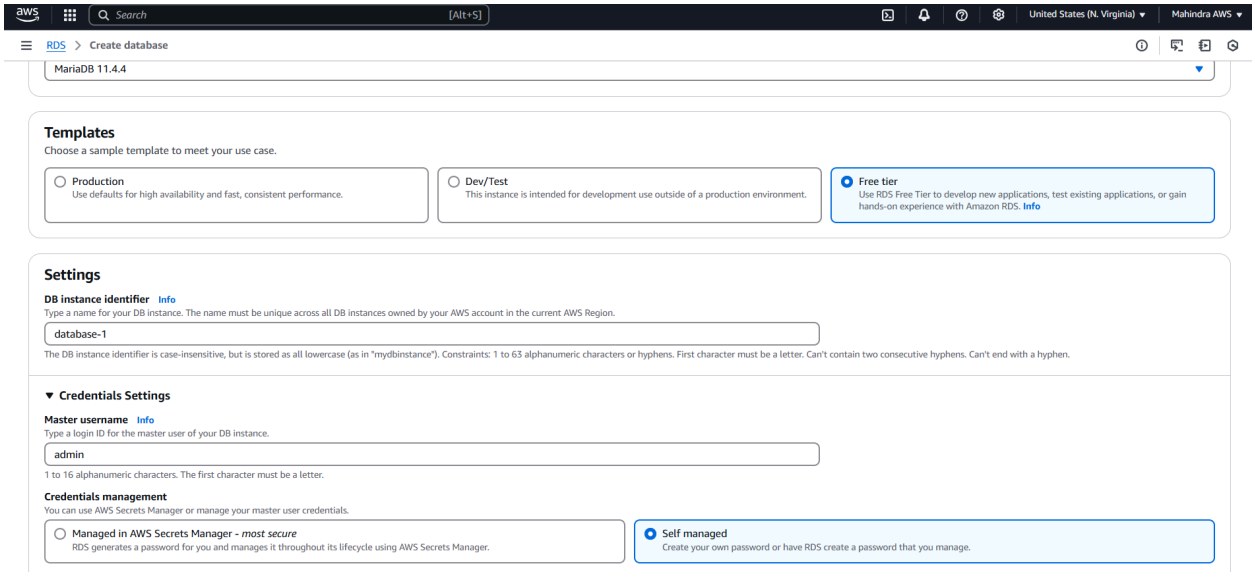
3. Under Choose A Database Creation Method, select Standard Create.



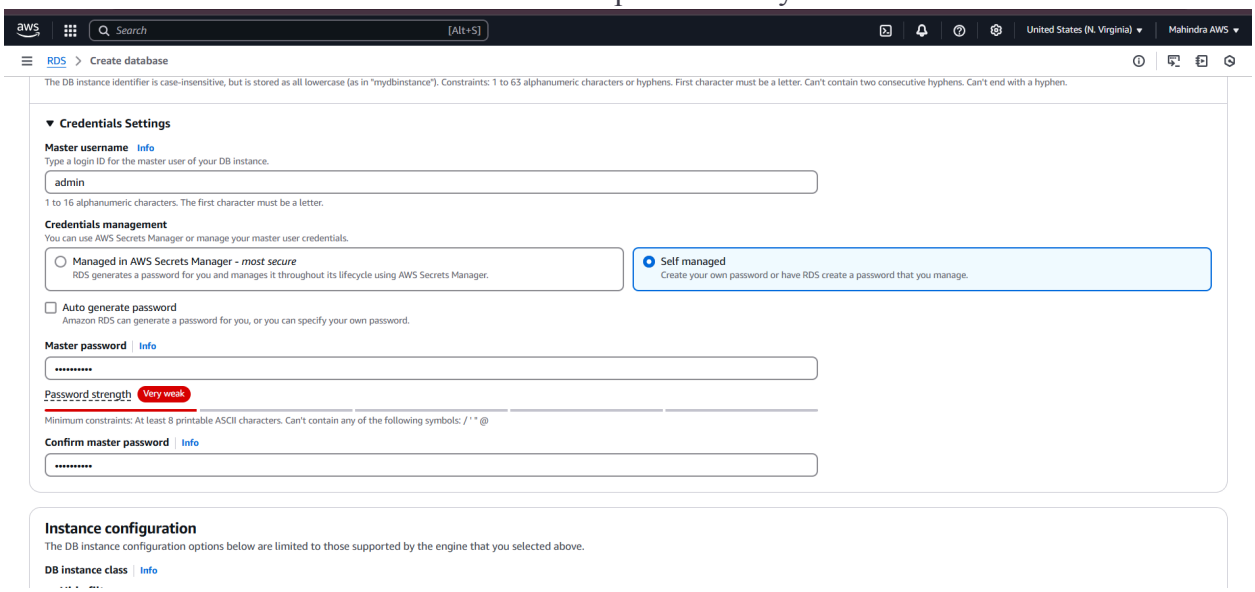
4. Under Engine Options, select MariaDB. Keep the default version.



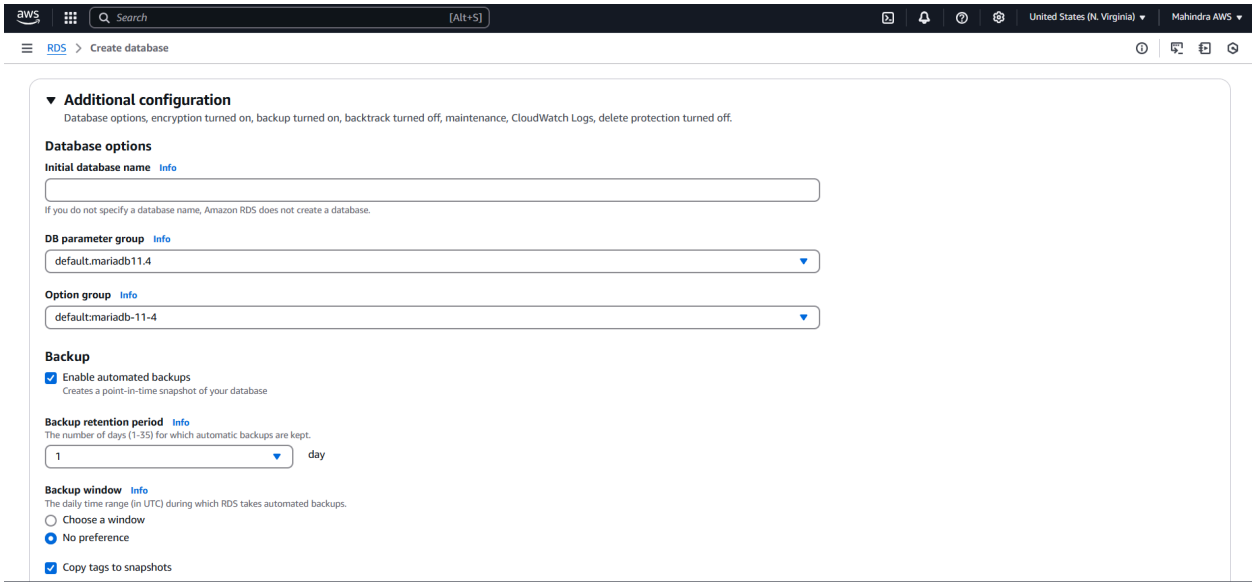
5. Under Templates, select Free Tier.



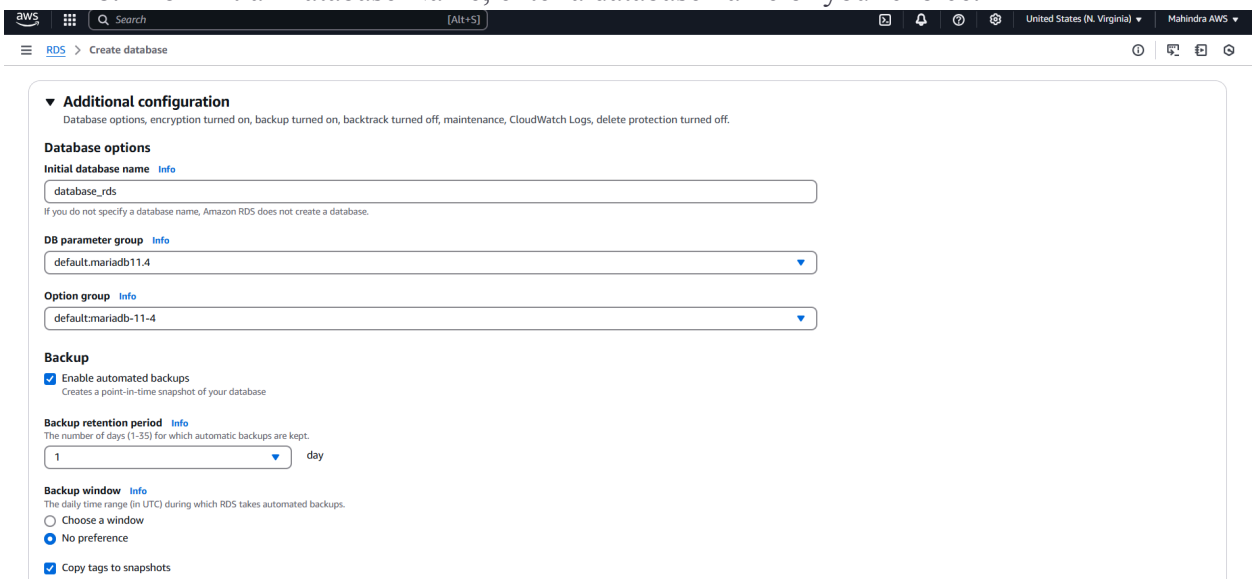
6. Enter a master username and master password of your choice.



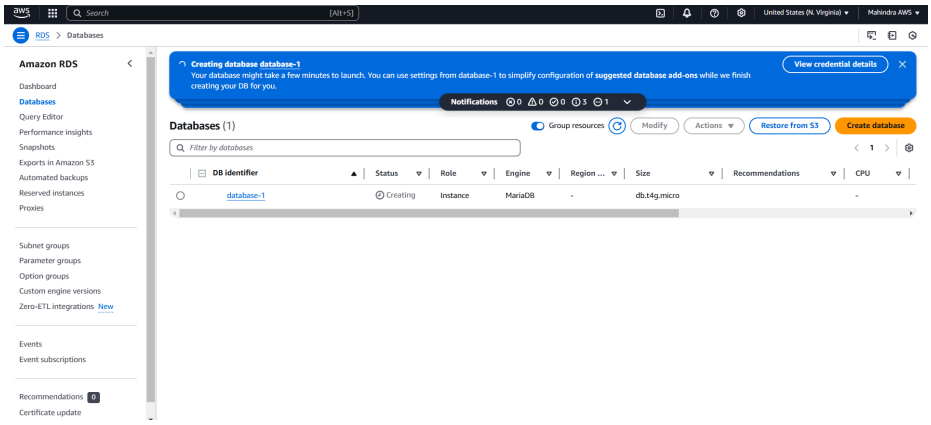
7. Scroll down to the Additional Configuration section and expand it.



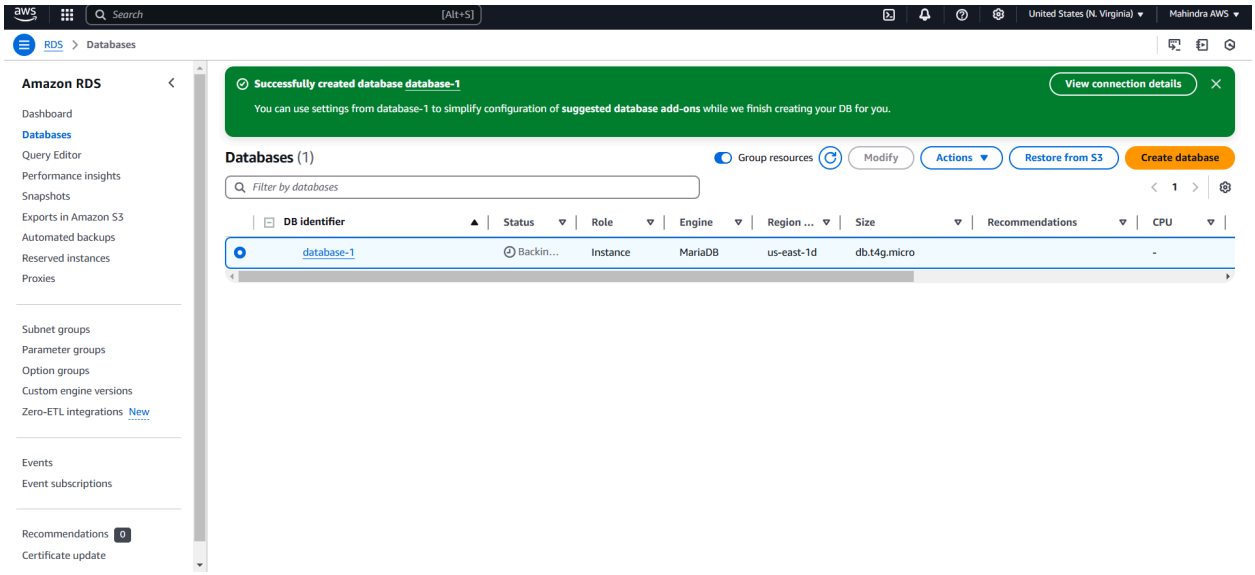
8. For Initial Database Name, enter a database name of your choice.



9. Click Create Database.



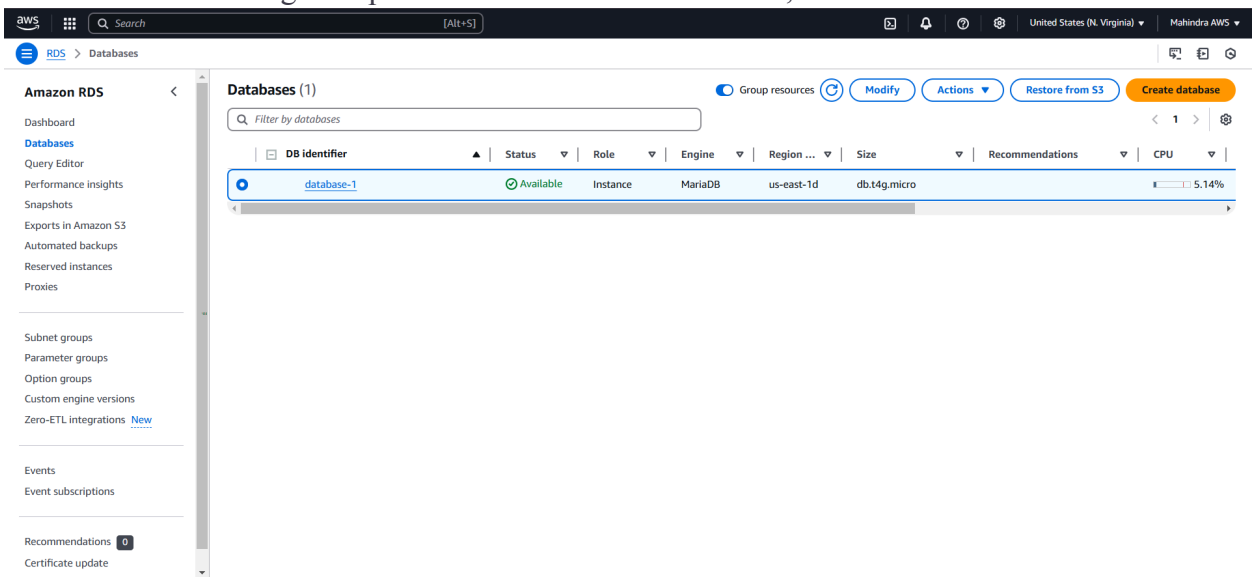
Wait for few mins. Then your database would have been created.



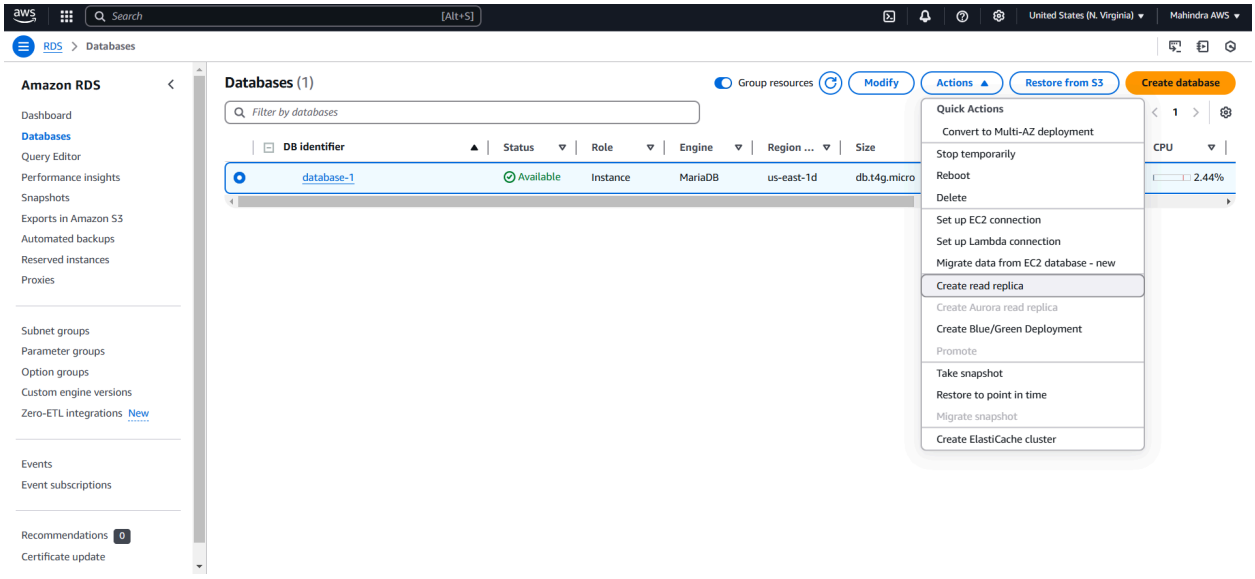
EXERCISE 5.2

Create a Read Replica

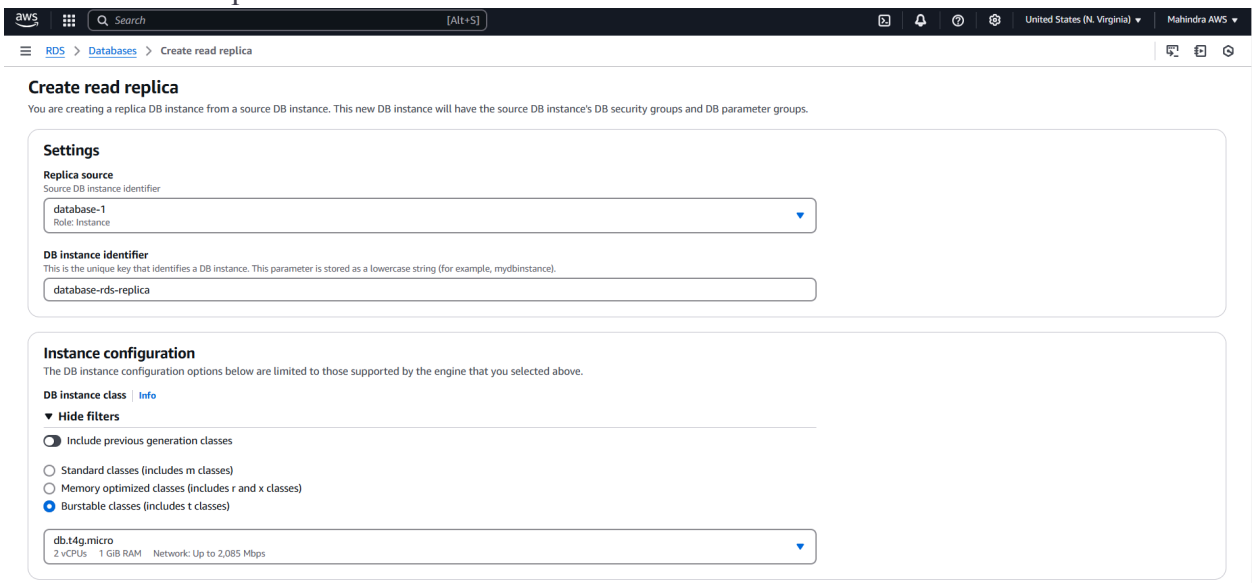
1. In the navigation pane of the RDS service console, click Databases.



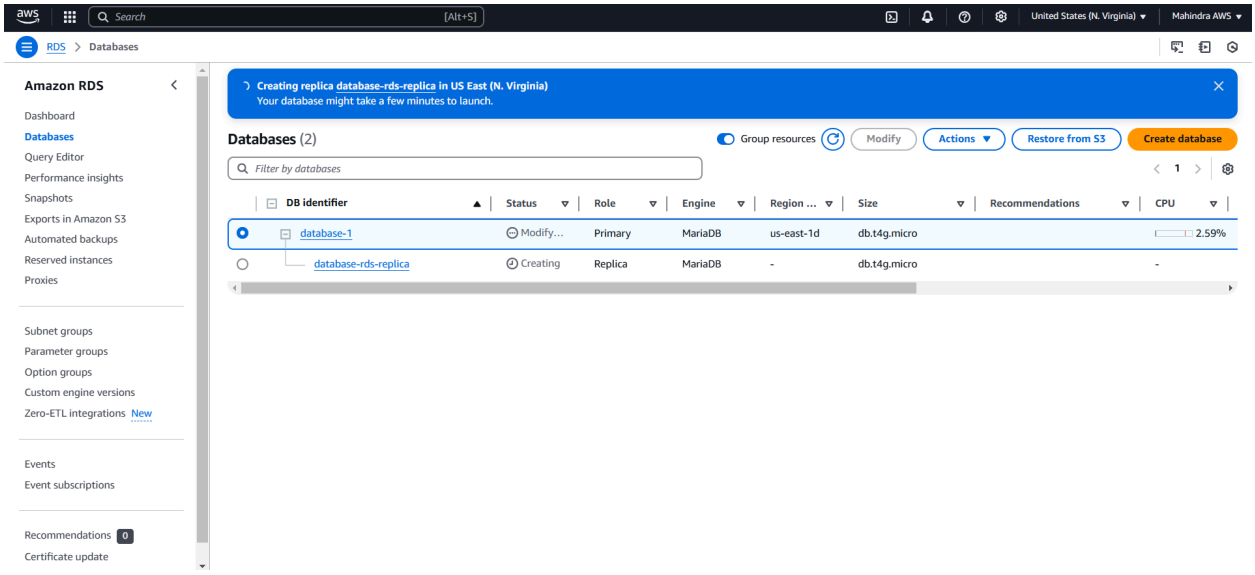
2. Select the instance you created earlier, click Actions, and then click Create Read Replica.



3. Under the Settings section, in the DB Instance Identifier field, enter a name for your read replica.



4. Click the Create Read Replicas button.

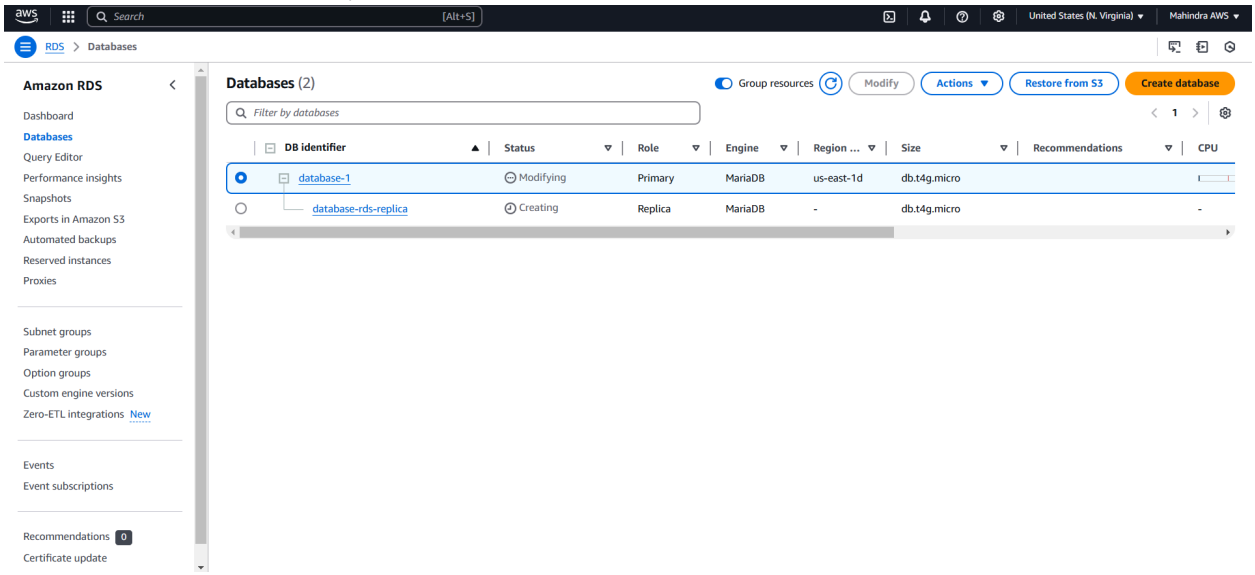


EXERCISE 5.3

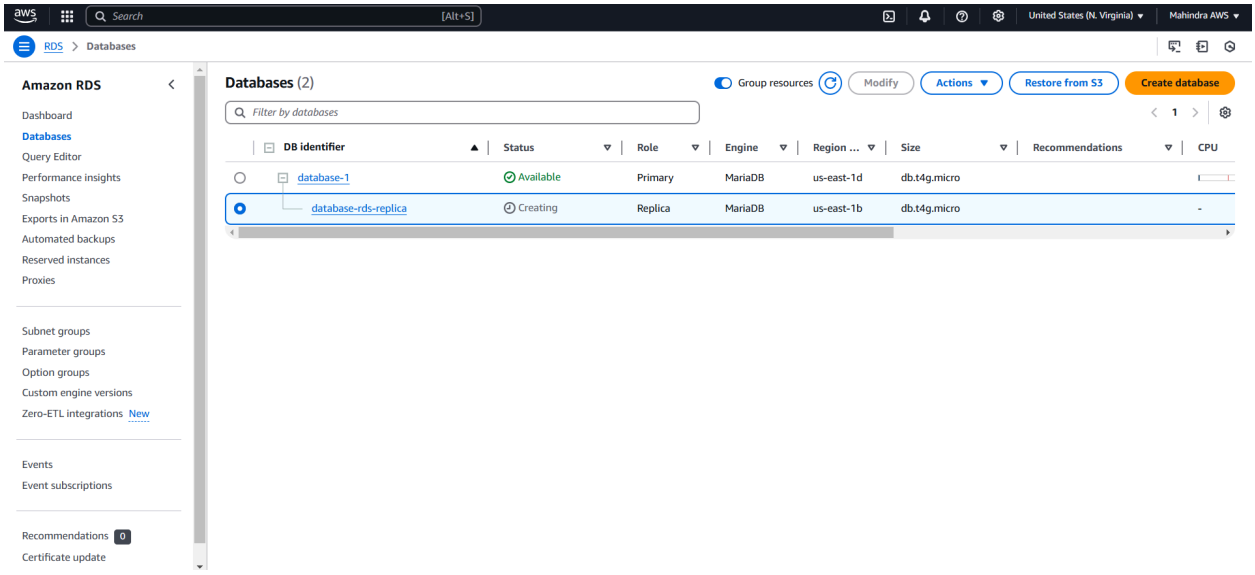
PROMOTE THE READ REPLICA TO A MASTER

In this exercise, you'll promote the read replica you just created to master. Before beginning, wait for the replica to enter the Available state.

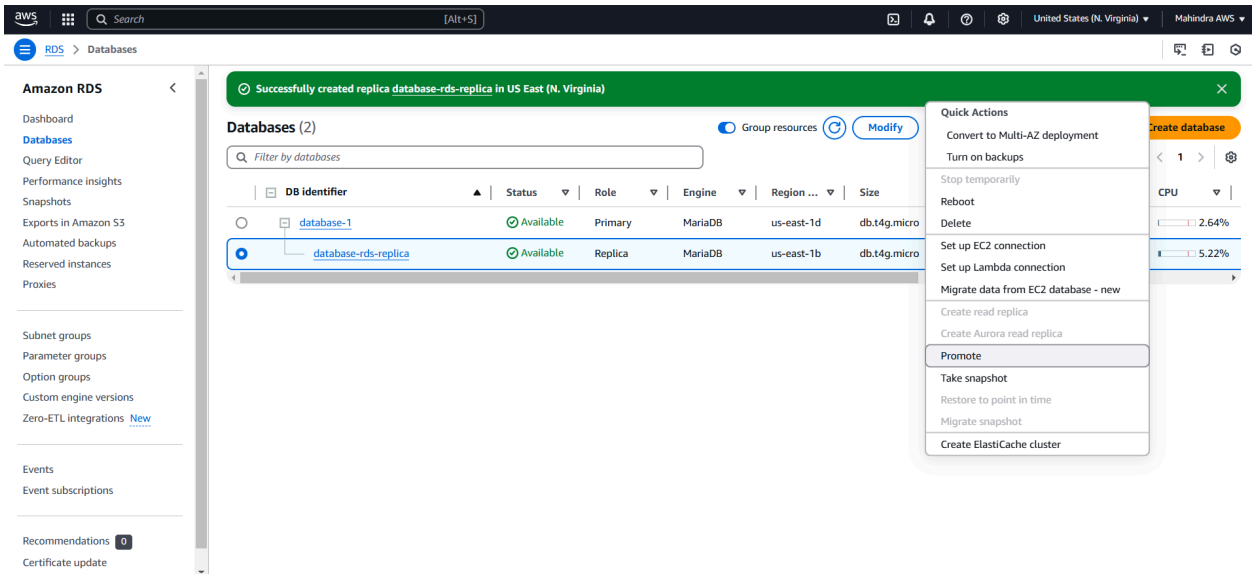
In the RDS service console, click Databases.



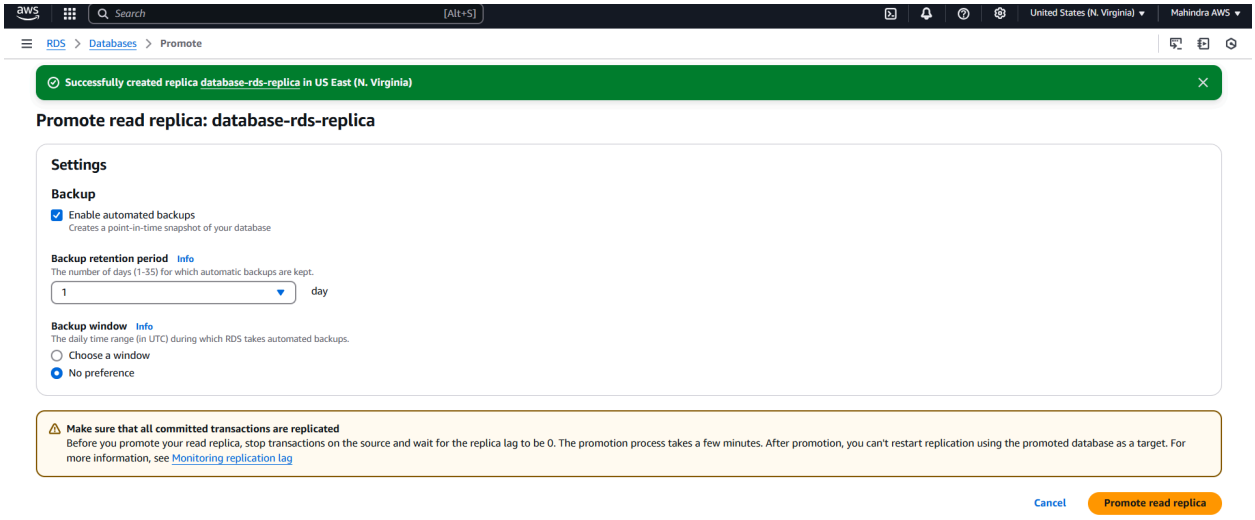
Select the read replica you created.



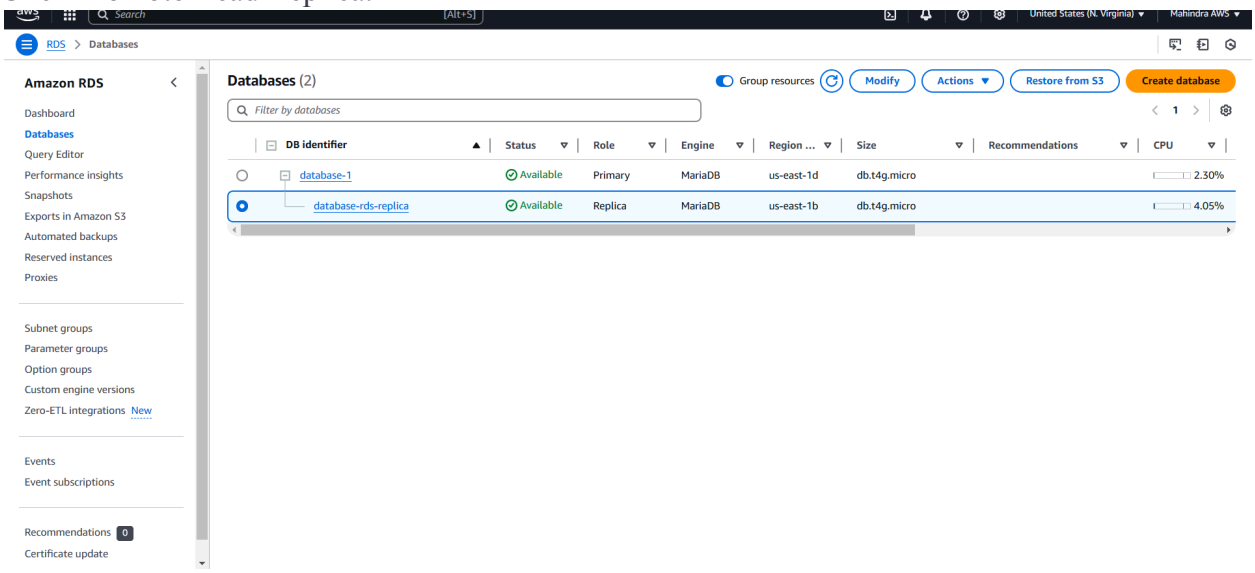
Click Actions and then select Promote.



Click Continue.



Click Promote Read Replica.



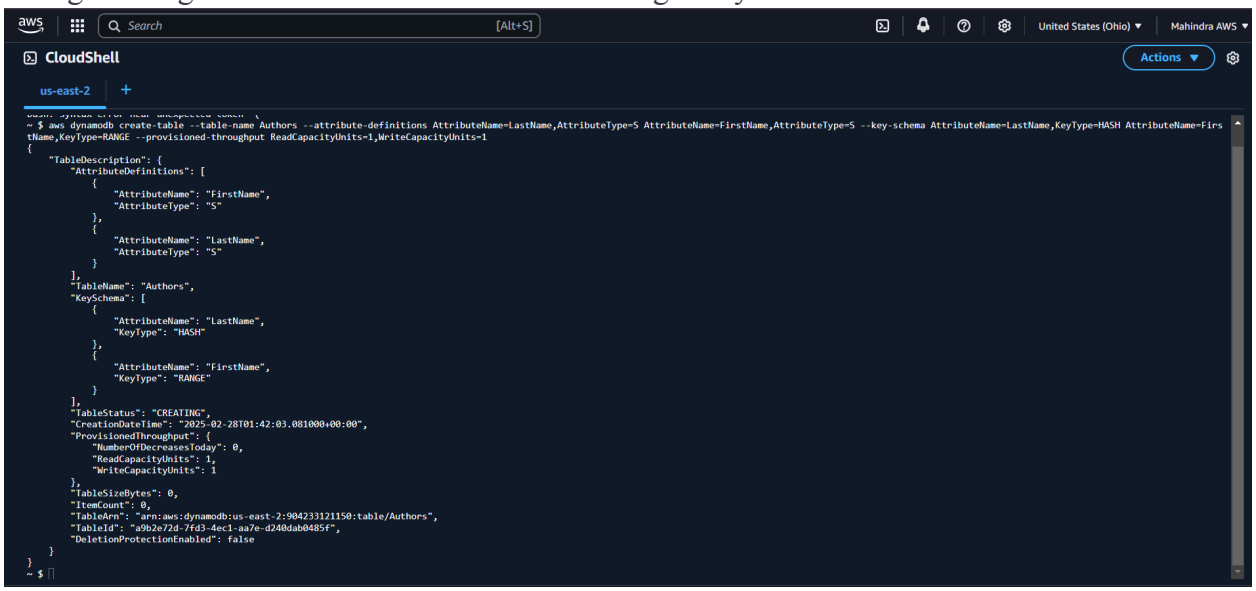
EXERCISE 5.4

CREATE A TABLE IN DYNAMODB USING PROVISIONED MODE

Use the following command to create a table named Authors with a partition key named LastName and sort key named FirstName. Both keys should use the string data type. Provision the table with a WCU and an RCU of 1.

```
aws dynamodb create-table —table-name Authors —attribute-definitions
AttributeName=LastName,AttributeType=S
AttributeName=FirstName,AttributeType=S
—keyschema AttributeName=LastName,KeyType=HASH
AttributeName=FirstName,KeyType=RANGE
—provisioned-throughput ReadCapacityUnits=1,WriteCapacityUnits=1
```

Change the region to Ohio because it is standard region by default



Go to the DynamoDB service console to view the table you just created.

